

L'irréductibilité de l'espace des courbes de genre donné

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Fixons un corps algébriquement clos k . Soit M_g be the space of modules of curves of genus g over k . Le résultat principal de cette note est que M_g est irréductible pour tout k . Bien entendu, le fait que M_g is irréductible ne dépend que de la caractéristique of k . When la caractéristique is 0, nous pouvons supposer that $k=\mathbb{C}$, and then le résultat est classique. Une preuve simple se trouve dans Enriques-Chisini [E, vol. 3, chap. 3], fondée sur l'analyse de the totality of coverings of P^1 of degree n , with a fixed number d of ordinary branch points. Cette méthode a été étendue to char. p by William Fulton [F], en utilisant des spécialisations from char. 0 to char. p provided that $p > 2g+1$. Malheureusement, les tentatives pour étendre cette méthode to all p semblent buter sur des questions difficiles of de ramification sauvage. Aujourd'hui, the Teichmüller theory donne une compréhension entièrement analytique mais très profonde into this irreducibility when $k=\mathbb{C}$. Notre approche est cependant la plus proche de Severi's preuve incomplète ([Se], Anhang F; the error is on pp. 344-345 and seems to be quite basic) and suit une suggestion of Grothendieck for using the result in char. 0 pour déduire the result in char. p . La base de both Severi's and Grothendieck's ideas est de construire des familles de courbes X , some singular, with $pa(X)=g$, over non-singular parameter spaces, which in some sense contain enough courbes singulières to link together any two composantes that M_g might have. Le point essentiel that fait fonctionner cette méthode now is a recent " théorème de réduction stable " for variétés abéliennes. This result fut d'abord démontré indépendamment in char. 0 by Grothendieck, au moyen de méthodes de cohomologie étale (private correspondence with J. Tate), and by Mumford, applying the easy half of Théorème (2.5), to go from curves to variétés abéliennes (cf. [M2]). Grothendieck a récemment renforcé his method de sorte qu'il s'applique en toute caractéristique (SGA 7, 1968) 9 Mumford has also given a proof using theta functions in char. ~ 2 . Le résultat est le suivant:

Théorème de réduction stable. — Soit R soit un anneau de valuation discrète with corps des fractions K .

Soit A soit un abelian variety over K . Then there exists a extension algébrique finie L of K such (x) The first author wishes to thank the Institut des Hautes]-tudes scientifiques, Bures-sur-Yvette, for support in this reseaπh and N. KATZ, for his invaluable assistance in the preparation of this manuscript; the second author wishes to thank the Tata Institute of Fundamental Reseaπh, Bombay, and the Institut des Hautes]-tudes scientifiques.

that, if $R \subset L =$ clôture intégrale of R in L , and if $di_{\sim}$ is the modèle de Néron of $A \times_{\sim} L$ over R , then the closed fibre $AL_{\sim}$ of $di_{\sim}$ has no radical unipotent. Nous donnerons deux preuves apparentées of our résultat principal. One of these is quite elementary, and follows by quite techniques standard once the Théorème de réduction stable for variétés abéliennes is applied, in w 2, pour déduire an analogous théorème de réduction stable pour les courbes. L'autre preuve is plus puissante, and is based on the use of a catégorie plus grande than the catégorie des schémas, and on proving for the objets de cette catégorie many of the théorèmes standard for schémas, especially the Enriques-Zariski théorème de connexité (EGA 3, (4-3)). Malheureusement, this catégorie plus grande is pas tout à fait une catégorie — it is a simple type of 2-catégorie; in fact, if X, Y are objets, then $H^0m(X, Y)$ is itself a catégorie, but one in which all morphismes are isomorphismes. The objets of this 2-catégorie we call champs algébriques (1). The espace de modules M_g is just the " variété grossière sous-jacente " of a more fundamental objet, the champ des modules =gggstudied in [Ma]. Les détails complets on the basic proprieties and theorems for champs algébriques seront donnés ailleurs. In this paper, nous ne donnerons que définitions and state sans preuve the general theorems which we apply. En utilisant la méthode of champs algébriques, nous pouvons démontrer non seulement the irreducibility of M_g itself, mais aussi de tous higher level espace de modules of curves too (cf. § 5 below).

1. Courbes stables et leurs modules

La définition clé de tout l'article is this:

Définition (x.x). — Soit S soit unny schéma. Soit $g > 2$. A stable curve of genus g

over S is a morphism $\pi: C \rightarrow S$ propre et plat whose fibres géométriques are réduites, connexes, 1-dimensional schémas C_s such that: (i) C_s has only ordinary doublepoints; (ii) if E is a composante rationnelle non singulière of C_s , then E meets the other composantes of C_s in more than 2 points; (iii) $\dim H^1(O_{C_s}) = g$. Nous étudierons dans cette section trois aspects of la théorie des stable curves: their pluri-canonical systèmes linéaires, their déformations, and their automorphismes. Supposons que $\pi: C \rightarrow S$ is a stable curve. Puisque π is plat and its fibres géométriques are intersections complètes locales, the morphisme π is localement a complete intersection (i.e., localement, C is isomorphe as S -schéma to $V(f_1, \dots, f_n)$ where $U \subset S$ is open, and $f_1, \dots, f_n$ are a regular sequence). Par conséquent, by la théorie des dualité of faisceaux cohérents [H], there is a faisceau inversible canonique $\omega_{C/S}$ on C — the unique groupe de cohomologie non nul of the complex of sheaves $f^*(0_S)$. Nous aurons besoin de savoir the faits suivants about $\omega_{C/S}$: a) pour tous les morphismes $f: T \rightarrow S$, $\omega_{C_T/T}$ is canonically isomorphe to $f^*(\omega_{C/S})$; (1) A slightly less general catégorie of objets, called algebraic spaces, has been introduced and studied very deeply by M. ARTIN [Ai] and D. KUMMERS [K]. The idea of enlarging the catégorie of varieties for the study of espace de modules is due originally, we believe, to A. WEIL.

b) if $S = \text{Spec}(k)$, k algébriquement clos, let $f: C' \rightarrow C$ be the normalisation of C , $x_1, \dots, x_n, y_1, \dots, y_r$ the points of C' such that the $z_i = f(x_i) = f(y_i)$, $1 \leq i \leq n$, are the points doubles of C . Then $\omega_{C/S}$ is the sheaf of i -forms on C' régulière sauf pour des pôles simples at the x 's and y 's and with $\text{Res}_i(\omega) + \text{Re}_i(\omega) = 0$; c) if $S = \text{Spec}(k)$, and $\mathcal{O}$ is a coherent sheaf on C , then $H^0(m, H^1(C, \mathcal{O} \otimes \omega_{C/S})) = H^0(m, \mathcal{O} \otimes \omega_{C/S})$.

Théorème (x.2). — If $g > 2$ and C is a stable curve of genus g over an algébriquement clos

corps k , then $H^1(C, \omega_{C/k}) = 0$ if $n > 2$, and $\mathcal{O}$ is very ample if $n > 3$. Démonstration. — Puisque C is stable, of genus $g > 2$, every irréductible component E of C either 1) has (arithmetic) genus > 2 itself, 2) has genus 1, but meets other composantes of C in at least one point, or 3) is non-singular, rational and meets other composantes of C in at least three points. But by b) above, $\omega_{E/k}$ is isomorphe to $\omega_{E/k}(-\sum Q_i)$, where $\{Q_i\}$ are the points where E meets the rest of C . Puisque the degree of $\omega_{E/k}$ is $2g_E - 2$, it follows that in any of the cases 1, 2 or 3, $\omega_{E/k}(\sum Q_i)$ has positive degree. This shows immediately that $\omega_{C/k}$ is ample on each component E of C , hence is ample. $H^1(C, \omega_{C/k}) = 0$. Puisque $\mathcal{O}_E$ has positive degree, by c) above, $H^1(\omega_{C/k})$ is dual to $\omega_{C/k} \otimes \omega_{C/k}^* = \mathcal{O}$. $H^1(\omega_{C/k}) = 0$ if $n > 2$, and so $H^1(\omega_{C/k}) = 0$ if $n > 2$. To prove that an invertible sheaf $\mathcal{L}$ on any schéma C , propre over k , is very ample, it suffices to show a) for all point fermés $x \in C$, $H^0(C, \mathcal{L}(x)) \rightarrow H^0(x, \mathcal{L}(x))$ is surjectif, b) for all point fermés $x, y \in C$, $H^0(C, \mathcal{L}(x+y)) \rightarrow H^0(x+y, \mathcal{L}(x+y))$ is surjectif. Using the exact sequence of cohomology, these both follow if $H^1(C, \mathcal{L}) = 0$. $H^1(C, \mathcal{L}) = 0$ if $n > 2$. In our case, $\omega_{C/k}$ is very ample, so if we use duality, we must show: (*) $H^0(m, H^1(C, \omega_{C/k} \otimes \mathcal{L})) = 0$, if $n > 2$. If x is a non-singular point, $\mathcal{L}_x$ is an invertible sheaf. If x is a double point, let $\pi: C' \rightarrow C$ be the result of blowing up x , and let $x_1, x_2 \in C'$ be the two points in $\pi^{-1}(x)$. Then it is easy to check that for any invertible sheaf $\mathcal{L}$ on C : $H^0(m, H^1(C, \mathcal{L})) = H^0(m, H^1(C', \pi^*\mathcal{L}))$. Par conséquent, we have 3 cases of (*): Case 1. — x, y non-singular points of C , then H^0 if $n > 2$.

Case 2. — x double point of C , $\pi: C' \rightarrow C$ blowing up x , $\{x_1, x_2\} = \pi^{-1}(x)$, and y a non-singular point of C . Then $H^0 = 0$, H^0 if $n > 2$. Case 3. — x, y points doubles of C , $\pi: C' \rightarrow C$ blowing

up x and y . $H^0 = (0)$ if $n > 2$. Then Now since the degree of $\mathcal{O}_C(n)$ on all components E of C is less than or equal to -2 , all of this is clear, except in those cases where $n = 2$, the degree of ω_C on some E is 1 , and in which two poles are allowed on E . This occurs if: (i) case 1, $\text{pa}(E) = 1$, E meets $C - E$ at only one point, $x, y \in E$. (ii) case 1, $\text{pa}(E) = 0$, E meets $C - E$ at only three points, $x, y \in E$. (iii) case 2, E a rational curve with one double point meeting $C - E$ at one point, $x = \text{double point of } E$. But in all these cases, C has components besides E and a section in the H^0 in question must definitely vanish on all these other components. So at the points where E meets $C - E$, the section has extra zeroes. Puisque the sheaf in question has degree 0 on E , the section is zero on E too. Q.E.D. Corollaire. — Soit $\sim : C \rightarrow S$ soit un stable curve of genus $g \geq 1$. Then $\sim^* \mathcal{O}_S(1)$ is relatively very ample if $n \geq 3$ and $\sim^* \mathcal{O}_S(1)$ is a locally free sheaf on S of rank $(2n-1)(g-1)$. Démonstration. — In fact, since for all $s \in S$, $H^1(\mathcal{O}_{C_s}(-n)) = 0$ it follows from [EGA, chap. 3, w 7], that $\mathcal{R}^1 L(\mathcal{O}_{C_s}(-n))$ is locally free and that $\mathcal{H}^0(\sim^* \mathcal{O}_S(1)) = 0$. Par conséquent the corollary follows. Q.E.D. Taking $n = 3$, it follows that every stable curve C/S can be realized as a family of curves in $\mathbb{P}^6$ with Hilbert polynomial: $P_C(n) = (6n-3)(n-3)$. ([MI], P. 99), it is easy to prove that there is a following standard arguments subschéma H^0 of $\mathcal{I}^1$ of all tri-canonically embedded stable curves. (H^0 is the Hilbert schéma over $\mathbb{Z}$.) To be precise, there is an isomorphism of foncteurs: $\mathcal{I}^1$ set of stable curves $\sim : C \rightarrow S$, plus isomorphismes: $H^0(\mathcal{O}_S(1), H^0(\sim^* \mathcal{O}_S(1))) \cong P(\mathcal{R}^1 L(\sim^* \mathcal{O}_S(-n))) \cong \mathbb{P}^6 \times S$ (modulo isomorphism) We will denote by Z_g the $H^0 \times \mathbb{P}^6$ the universal tri-canonically embedded stable curve. Le foncteur of stable curves itself is the sheafification of the quotient of foncteurs: $H^0/\text{PGL}(5g-6)$.

We now consider the deformation theory of stable curves. Soit k soit un ground corps. The deformation theory of X 's lisse over k can be found in [SGA, 6o-6I]; for singular S 's, the theory has been worked out in [Sc]. We shall indicate here the results of this theory for a schéma X which is (i) one-dimensional; (ii) generically lisse over k ; (iii) localement a complete intersection. The advantages of this case are two-fold: first, the "cotangent complex" of Grothendieck, Lichtenbaum and Schlessinger reduces, in view of (ii) and (iii), to the single coherent sheaf $f^* \Omega_{X/k}$, the Kahler differentials. Secondly, we have:

Lemme (I. 3). — $\text{Ext}^2(f^* \Omega_{X/k}, (g)_* \mathcal{O}_X) = (0)$.

Démonstration. — Use the spectral sequence: $H^p(X, \text{Ext}^q(f^* \Omega_{X/k}, \mathcal{O}_X)) \rightarrow \text{Ext}^{p+q}(\mathcal{O}_X, (g)_* \mathcal{O}_X)$. Then (i) $H^0(X, \text{Ext}^1(f^* \Omega_{X/k}, \mathcal{O}_X)) = (0)$ since $\dim X = 1$. (ii) Puisque $\Omega_{X/k}$ is locally free except at a fini number of points, $\text{Ext}^1(f^* \Omega_{X/k}, \mathcal{O}_X)$ has 0 -dimensional support, hence $H^1(X, \text{Ext}^1(f^* \Omega_{X/k}, \mathcal{O}_X)) = (0)$. (iii) Locally, if we embed $X \subset \mathbb{A}^n$, then $f^* \Omega_{X/k}$ has a free resolution of length 2 : $\mathcal{O}_X \rightarrow \mathcal{O}_X^{\oplus n} \rightarrow \mathcal{O}_X^{\oplus m} \rightarrow f^* \Omega_{X/k} \rightarrow 0$ where $\mathcal{O}_X^{\oplus m}$ = sheaf of ideals defining X . Par conséquent $\text{Ext}^2 = (0)$. Q.E.D. In Schlessinger's theory, the significance of Lemme (1.3) is that all obstructions vanish, i.e., déformations of X over base schémas $\text{Spec}(A/U)$ (A = local Artin ring with residue corps k) can always be embedded in déformations over $\text{Spec}(A)$. Moreover, the theory says that there is a canonical one-one correspondence between $\text{Ext}^1(f^* \Omega_{X/k}, (g)_* \mathcal{O}_X)$ and the first order déformations of X , i.e., propre, plat morphismes p , and isomorphismes as follows

Let $\mathcal{X}$ be a deformation of X over $\text{Spec } k[[\varepsilon]]$, proper, flat morphism p , with section

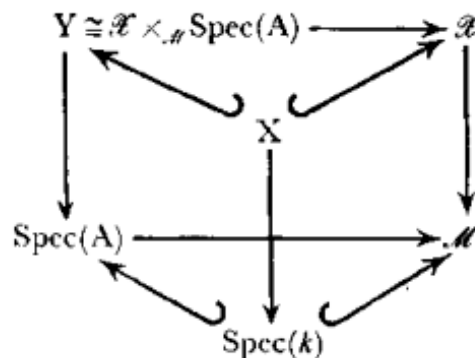
$$\begin{array}{ccccc} X_1 & \supset & X_1 \times_{\text{Spec } k[[\varepsilon]]} \text{Spec } k & \xleftarrow{\sim} & X \\ \downarrow p & & \downarrow & & \downarrow \\ \text{Spec } k[[\varepsilon]]/\varepsilon^2 & \supset & \text{Spec } k & & = \text{Spec } k. \end{array}$$

obstructions vanish there is a normal formal deformation $\mathcal{X}$ of X over

: $X \sim D \times_k \text{Spec}(k) \hookrightarrow X$ ($x \in \text{Spec}(k) \mapsto D_x = \text{Spec}(k)$). Puisque les obstructions
 vanish, there is a versal formal deformation $\tilde{X}$ of X over the base schéma $\tilde{S} = \text{Spec}$
 $\mathcal{O}_k[[t_1, \dots, t_n]]$, where $\mathcal{O}_k = k$ if the char. is 0, or the complete regular local ring, max.
 ideal $\mathfrak{p}$. $\mathcal{O}_k$, residue corps k , if char. $(k) = p$, unique (by Cohen's structure theorem)
 and $N = \dim_k \text{Ext}^1(\Omega_X, \mathcal{O}_X)$

So P. DELIGNE AND D. MUMFORD This means that Y is a formal schéma, propre and plat
 over M , with fibre X over $\text{Spec}(k)$ and the two properties: a) Every deformation of X is induced
 from Y , i.e., if A is a local Artin $\mathcal{O}_k$ -algebra with residue corps k , and $p: Y \rightarrow \text{Spec}(A)$ is propre
 and plat with fibre X over $\text{Spec}(k)$, then there is a commutative diagram:

commutative diagram:



$\mathfrak{t}[\epsilon]/(\epsilon^2)$, the above morphism f is uniquely determined

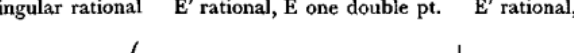
$Y = \tilde{X} \times_{\text{Spec}(A)} \text{Spec}(A) \rightarrow \text{Spec}(A) \rightarrow \text{Spec}(k)$ b) If $A = k[\epsilon]/(\epsilon^2)$, the above morphism f is
 uniquely determined by the diagram. This implies that the tangent space to Def_X at its point
 fermé is canonically isomorphic to $\text{Ext}^1(\mathcal{O}_X/k, \mathcal{O}_X)$. In case $\text{Ext}^1(\mathcal{O}_X/k, \mathcal{O}_X) = (0)$, $\tilde{X}$ is, in fact,
 universal: i.e., in property a), f is always unique, which means that the foncteur represented by
 $\tilde{X}$ in the catégorie of artin, local $\mathcal{O}_k$ -algebras is isomorphic to the foncteur of déformations Y/A of
 X . This fortunately holds for stable curves:

Lemme (I.t). — If X is a stable curve, $\text{Ext}^1(\mathcal{O}_X/k, \mathcal{O}_X) = (0)$.

Démonstration. — We may assume that k is algébriquement clos. Now a homo-
 morphism $f: \tilde{X} \rightarrow \text{Spec}(k)$ is given by an everywhere regular vector corps D
 on X . Such a vector corps is given, in turn, by a regular vector corps D' on the
 normalisation X' of X which vanishes at all points of X' lying over the points doubles
 of X . In particular, D' and hence D vanishes identically on all composantes E of X
 whose normalisation E' has genus > 2 . There remain the following possibilities for E :



E non-singular rational E' rational, E one double pt. E' rational, $E \geq 2$ double pts.



E non-singular elliptic E' elliptic, $E \geq 1$ double points

In all cases where E' is rational, note that D' has to have at least 3 zeroes; and where E' is elliptic, D' has to have at least one zero. So D' vanishes on all components E' . This proves that $\text{Ext}^1_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X) = 0$. Q.E.D. Schlessinger's theory also allows us to trace what happens to the singularities of X in this deformation. For each point $x \in X$, he studies deformations of the complete local ring $\hat{\mathcal{O}}_{X,x}$ alone, i.e., flat A -algebras $\hat{\mathcal{O}}_{X,x}$ plus isomorphisms: $0 \rightarrow \Omega_{X,x}$.

are classified by $\text{Ext}^1(\mathcal{F}_i, \mathcal{O}_X)$. And there is a versal formal deformation $\tilde{\cdot}$, which is a complete local ring and a $\mathcal{O}_X[[t_1, \dots, t_N]]$ -algebra, $N = \dim_k \text{Ext}^1(\mathcal{F}_i, \mathcal{O}_X)$, such that $\tilde{g}/(\tilde{p}, \tilde{h}, \dots, \tilde{t}_N) = \tilde{g}, x$. Whenever X is lisse over k at x , $N = 0$, and the theory is uninteresting. The first non-trivial example is a k -rational ordinary double point with k -rational tangent lines: $\mathcal{O}_X = k[u, v]/(u^2 - v^2)$. Then $N=2$, and $\tilde{g} = k[u, v]$. In other words, if 0 is any deformation of $\mathcal{O}_X$, x and u, v are lifted suitably into $\tilde{g}$, then $u - v = w \in \mathfrak{m}_A$ and 0 is induced from $\tilde{\cdot}$ via the homomorphism $\mathcal{O}_X[[t]] \rightarrow A$ taking t_i to w . This is easy to prove. Finally, Schlessinger's theory connects global deformations to local ones. Soit $\tilde{\cdot}/\sim, \tilde{g}$ with $\tilde{\cdot}/\sim = \text{Spec}(A)$ be the versal global deformation, let $x_1, \dots, x_n \in X$ be the points where X is not lisse over k , and let $\tilde{\cdot}_i$ as A_i -algebra be the versal deformation of the local ring $\tilde{\cdot}_i, x_i$. Soit $\tilde{\cdot} \rightarrow \tilde{g}, 0 = \text{Spec}(A) \rightarrow \text{Spec}(A_i)$. Then we may consider the local rings of $\tilde{\cdot}$ at x_i . There are $\mathcal{O}_X$ -homomorphisms $\varphi_i: A/\mathfrak{m}_i \rightarrow A$ such that $\mathcal{O}_{i, x_i} \cong \mathcal{O}_i|_A$. Dualizing, we obtain a morphism $\mathcal{O} = H^1(\text{Spec}(\varphi_i)) : \text{dt}'_{\text{gl}} \rightarrow \text{dt}'_{\sim}$ which describes exactly how the various singularities of X behave in the versal deformation $\tilde{f}$. The final fact that we need is:

Proposition (T. 5)- — $\varphi : \sim'gz \longrightarrow \sim'z0$ is formally lisse, i.e., there are isomorphisms

$$-g_{i0} \sim \text{Spec } o[[t_1, \dots, t_N]] \text{ such that } \sim^*(t_i) = t_i, 1 \leq i \leq N.$$

5

formal deformation $\tilde{f}/M_r$ where $\dots o, [[t_1, \dots, t_n]]$. Note that since in this case, the invertible sheaf $\tilde{c}/\tilde{\sim}$ is relatively ample, $\tilde{\sim}$ is not seulement a formal schéma over $\tilde{M}_r'$, but also the formal completion of a unique schéma propre and plat over $\tilde{M}_r'$, which we will also denote by $\tilde{\sim}$. $\tilde{c}_g$ is clearly a stable curve over $\tilde{M}_r'$. Now each double point x_i has one modulus (cf. our example above) so the versal deformation space of the rings $O_{\tilde{\sim}, i, c}$ is $\tilde{o}[[t_1, \dots, t_n]]$. By Proposition (I.5), we may identify t_i with $t_{\tilde{\sim}}$, and we conclude that for suitable u, v : $\dots$, In particular, $t_r = o$ is the locus in $\tilde{\sim}$ where x_i remains a double point. The relation between the formal espace de modules $\tilde{M}_r'$ of C and the local structure of H_g at a point x with $\times$ corresponding to some tri-canonical model of C is exactly the same as in the case of non-courbes singulières ([M1], chap. 5, w 2). Soit $\tilde{\sim}$ be the completion of the local ring $O_{\tilde{\sim}, H_g}$, and let $T = \text{Spec}(\tilde{\sim})$. Soit x denote the point fermé of T too. The universal family of stable curves $Z_g \times H_g \times 5g-6$ induces a family $Z' \times T \times$ whose fibre Z' , over x is C . Then there is a unique morphisme $f: T \rightarrow$ such that $Z' \times$, with this isomorphism restricting to the identité on the fibres over x , both of which are C . I claim that via f , T is formally lisse over $\tilde{\sim}/t'$, i.e., $O_{\tilde{\sim}, \text{ok}}[[t_1, \dots, t_N, t_{N+1}, \dots, t_n]]$. In fact, by choosing an isomorphism $p \sim \sim | (\sim r \text{ Tphg-6X}^n f'$, we obtain a tri-canonical embedding $\tilde{c} \rightarrow \text{psg-6} \times 162$ of $\tilde{c}$, hence a morphisme $s: \tilde{\sim} \rightarrow H_g$ such that $\tilde{\sim}$, with this embedding, is the pull-back of Z_g . Then s factors through T and $s: \tilde{\sim} \rightarrow T$ is a section off. On the other hand, consider the action of $\text{PGL}(hg-6)$ on H_g . Soit $S_{\tilde{\sim}}$ be the stabilizer of the k -valued point x . Then $S_{\tilde{\sim}}$ is fini and réduites. Because if it were not, $S_{\tilde{\sim}}$ would have a non-trivial tangent space at the origin, i.e., there would soit un $k[z]/(z^2)$ -valued point of $\text{PGL}(hg-6)$ centered at the identité, which maps the embedded stable curve $C \subset \text{phg-} \sim$ corresponding to x into itself. But this action is given by an everywhere regular derivation on C , and we have

seen that all such vanish. This means that this $k[r]/(r)$ -valued automorphisme is the identité at all points of C and, since C is connexes and spans psg-6, the automorphisme is the identité everywhere. Thus $S_{\tilde{\sim}}$ is fini and réduites. It follows that the action of $\text{PGL}(5g-6)$ on T is formally free, and hence that T is formally a principal fibre bundle over M_r' with group $\text{PGL}(5g-6)$. Par conséquent T is formally lisse over $\tilde{W}$ as required. Putting this together with what we know about $\tilde{\sim}/t'$, we conclude the following: Soit k soit unny algébriquement clos corps, let $H'_g = H_g \times \text{Spec}(0k)$, $Z'_g = Z_g \times \text{Spec}(ok)$, let $x \in H'_g$ soit un point fermé, let $C \subset Z'_g$ be the stable curve over x , let $x_1, \dots, x_t \in C$ be its points doubles. Then

Théorème (I. 6). — There are isomorphismes

$$\text{ok}[[t_1, \dots, g_n, z; \sim \text{ok}[[u_i, v, t_x, \dots, t_i]]]/(u \sim v, -t,).$$

Corollaire (x. 7). — H_g is lisse over Z . In particular, for all algébriquement clos corps k ,

$H_g \times \text{Spec}(k)$ is a disjoint union of a fini number of non-singular algebraic varieties over k . Soit $0_{\{x \in H_g \mid \text{the corresponding stable curve } (Z_g)_{\tilde{\sim}} \text{ is non-singular}\}}$. $H_g - S = \{x \in Z_g \mid \text{the projection } 7: Z_g \rightarrow H_g \text{ is not lisse at } x\}$.

Définition (x . 8). — Soit $p: X \rightarrow Y$ soit un lisse morphisme offini type, with Y a noetherian

schéma, and let $D \subset X$ soit un relative Cartier divisor. Then D has normal crossings relative to Y if for all $x \in D$, the local equation $d = o$ of D decomposes in the strict completion (1) $\tilde{\sim}_x$ of $0 \sim x$ as $d = d_1 \dots d_k$, where $d_1, \dots, d_k$ are linearly independent in $\tilde{f}t_x, x/\tilde{f}t_{\tilde{\sim}}, x + my, y' \sim, x$, with $y = p(x)$.

Corollaire (x.9). — $H^0 *$, where S^* is a divisor with normal crossings relative

to Z . Z_g and S are lisse over Z , and the projection $p : S \rightarrow S^*$ is fini and an isomorphisme at all points where S^* is lisse over Z , i.e., S is the normalisation of S^* . **Proof** — In the notation of Théorème (1.6), S^* is defined in $g_{\sim, a}$ by the local equation $t_1 \dots t_k = 0$. And $\sim, Z_{\sim} \sim \text{Ok}[[U_i, V_i, t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_N]] \text{Ok}[[t_1, \dots, t_{\sim}, t_{+}, \dots, t_s]]$. **Q.E.D.** (x) The complete local ring, formally étale over $\sim, x$ with residue corps the separable closure of $d)x, x/\text{rex}, x$.

Next we take up the isomorphismes and automorphismes of stable curves. Supposons que $p : X \rightarrow S, q : Y \rightarrow S$ are two stable curves:

Définition (I. 10). — $\text{Isoms}(X, Y)$ is the foncteur on (Sch/S) associating to each S -schéma S'

the set of S' -isomorphismes between $X \times_S S'$ and $Y \times_S S'$. If $X=Y$, we denote $\text{Isoms}(X, X)$ by $\text{Aut}_S(X)$. Puisque both X and Y have the canonical polarizations $\text{CO}_X/s, \text{O}_Y/s$ respectively, any isomorphisme $f: X \rightarrow Y$ must satisfy $J^*(c_X/s) \sim \text{O}_X/s$. Par conséquent, by Grothendieck's results on the representability of the Hilbert schéma and related foncteurs [Gr1], we conclude that $\text{Isoms}(X, Y)$ is represented by a schéma $\text{Isoms}(X, Y)$, quasi-projective over S . Concerning this schéma, we have:

Théorème (x. II). — $\text{Isoms}(X, Y)$ is fini and unramified over S .

Démonstration. — To check that $\text{Isoms}(X, Y)$ is unramified, we may take S to be the spectrum of an algébriquement clos corps k , in which case $\text{Isoms}(X, Y)$ is either empty or isomorphe to $\text{Aut}_k(X)$. A point of $\text{Aut}_k(X)$ with values in $k[r, z]$ with image the identité may be identified with a vector corps on X . By Lemme (i.4), stable curves have no non-zero vector corps. This proves that $\text{Isoms}(X, Y)$ is unramified over S , and since it is also of fini type over S , it is quasi-fini over S . It remains to check that $\text{Isoms}(X, Y)$ is propre over S . Locally over S , X and Y are the pull-backs of the universal tri-canonically embedded stable curve by some morphismes from S to H_g , so that it suffices to prove the propreté of $\text{Isoms}(X, Y)$ in the "universal" case where $S = H_g \times H_g$, X and Y being the two image inverses of the universal curve on H_g . In that case, the ouvert of $\text{Isoms}(X, Y)$ corresponding to lisse curves is dense, so that the Théorème follows from the valuative criterion of propreté which holds by:

Lemme (I, I2). — Soit X and Y be two stable curves over a anneau de valuation discrète tl . with

algébriquement clos residue corps. Denote by $\sim$ and s the generic and point fermés of $\text{Spec}(P_{\sim})$, and assume that the fibre génériques $X_{\sim}$ and $Y_{\sim}$ of X and Y are lisse. Then any isomorphisme $\sim$ between $X_{\sim}$ and $Y_{\sim}$ extends to an isomorphisme $\sim$ between X and Y . (A posteriori, it follows from Théorème (I. I I) that the lemma holds for any valuation ring $P_{\sim}$, and without assuming X , or Y , lisse.) **Démonstration.** — Another way to put the lemma is that if we start with a lisse curve X , of genus $g > 2$ over the corps des fractions K of I , there is, up to canonical isomorphisme, at most one stable curve X over R with $X_{\sim}$ as its fibre générique. We shall deduce this from the analogous uniqueness assertion for minimal models ([L] and [S]): given a lisse curve $X_{\sim}$ of genus $g > 1$ over K , there is, up to canonical isomorphisme, at most one regular 2-dimensional schéma X , propre and plat over R , with $X_{\sim}$ as its fibre générique, without exceptional curves of the first kind in X . Soit z denote a generator of the maximal ideal of R and consider the affine plane curve C_n over R given by: $xy = z^n$.

Soit C . denote the schéma obtained by: 1) blowing up the maximal ideal at the unique singularity of C_n ; 2) blowing up the maximal ideal at the unique singularity of this schéma, and

so on [1] in S . It is easy to check that the regular whose fibre spéciale is the same as that of C_n except that the Singular point is replaced by a sequence of $n-1$ projective lines as follows:

1 projective lines as follows:



so x is a singular point of the stable curve X over R . $\square$

Now suppose x is a singular point of the stable curve X over R . At x , X is formally isomorphic as schéma over R to one of the schémas C_n , so we may blow up X the same way we blew up C_n . If we do this for all singular points of X , we get a regular schéma with fibre générique $X^\sim$. In addition, any composante rationnelle non singulière of X is linked to the other irréductible composantes by at least two points, hence it is not exceptional of first kind. Par conséquent X is the minimal model of $X^\sim$. Note finally that C_n is a normal schéma, hence so is X ; therefore X is the unique normal schéma obtained from $X^\sim$ by contracting all non-singular rational composantes of X linked to the other irréductible composantes by exactly two points. This proves that X is essentially unique. Q.E.D. Another important fact about the automorphisms of stable curves is:

Théorème (1.13). — Soit k soit un algébriquement clos corps and X a stable curve over k .

Soit $\text{Pic}^\sim$ denote the group of invertible sheaves on X of degree 0 on each component. Then the map (of ordinary groups) :

$\text{Aut}(X) \rightarrow \text{Aut}(\text{Pic}^\sim(X))$ is injective. **Démonstration.** — Soit φ soit un automorphisme of X inducing the identité on $\text{Pic}^\sim$

Lemme (I. 4). — If X is lisse, then $\sim$ is the identité.

Démonstration. — If not, by the Lefschetz-Weil fixed point formula, the number n of fixed points of φ counted with their multiplicities, is $n = 1 - \text{Tr}(\varphi^* \text{Tr}(\text{Pic}^\sim + 1)) = 9 - 2g < 0$ which is absurd. Q.E.D.

Lemme (x. 15). — If X is irréductible, then $\sim$ is the identité.

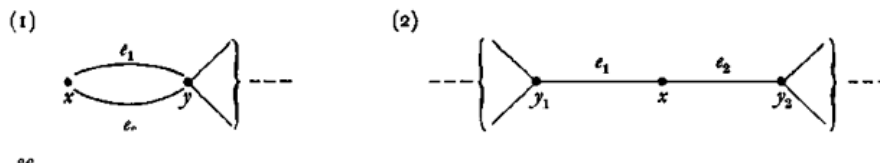
Démonstration. — Soit φ' be the action of φ on the normalisation X' of X . Each singular point of X , together with an ordering of its 2 image inverses in X' , defines a distinct morphisme from G_m to $\text{Pic}^\sim$ so that the image inverse S of the singular locus of X is pointwise fixed by φ' . One has either a) $\text{genus}(X') > 2$: then conclude by Lemme (I. 14); b) $\text{genus}(X') = 1$, $|S| > 2$, then φ' is a translation on X' leaving a point fixed, and so φ' is the identité; c) X' is the projective line, $|S| > 4$ and φ' is a projectivity leaving more than three points fixed, so is the identité. Q.E.D.

cible composantes of is the case, unless X is points. Q.E.D. Soit Γ be the following (unoriented) graph: (i) The set of vertices of Γ is the set Y_0 of irréductible composantes of X , (ii) the set of edges of Γ is the set pl of the singular points of X which lie on two distinct irréductible composantes, (iii) an edge e_i has for extremities the irréductible composantes on which e_i lies.

Lemme (x. i6). — If $\sim$ induces the identité on F , then $\sim$ is the identité.

Démonstration. — If X_1 is an irréductible component of X , then $9(X_i)=X_1$ and 9 leaves fixed the points of intersection of X_1 with the other composantes. In addition, $P_{Ie\sim}$ maps onto $P_{Ie\sim}$ so that φ acts trivially on $P_{Ie\sim}$. Either: a) genus $(X_1)>2$ and $9(X_i)$ is the identité by Lemme (I.15); b) genus $(X_1)=1$, φ acts by a translation and leaves a point fixed, so is the identité on X_1 ; c) X_1 is the projective line and φ leaves fixed at least three points, so is the identité on X_1 . Q.E.D. Lemme (I. I7). — (i) Any edge in F has distinct extremities. (ii) Any vertex which is the extremity of 0, 1 or 2 edges is fixed by 9 . (iii) $\sim$ acts trivially on $H^0(F, Z)$. Démonstration. — It is easy to check that the subgroup of $P_{Ic\sim}$ corresponding to invertible sheaves whose restriction to each irréductible component of X is trivial is canonically isomorphic to $H^0(F, Z)/m$. This implies (iii), and (i) is trivial. The morphisme from $P_{Ie\sim}$ to the product $O_{P_{Ie\sim}}$ extended over the irreducible X , is surjective, so that if $P_{Ie\sim} = \{e\}$, then $\varphi(X_i)=X_i$. This is a projective line, linked to the other composantes in at least three points. We prove now that if an automorphisme 9 of any finite graph P has the properties stated in Lemme (I. I7), it is the identité. Make induction on the sum of the number of vertices and edges of F . If P has an isolated point x , then $\varphi(x)=x$ so let $F^* = F - \{x\}$. Then $\varphi =$ identité on F^* by induction, so $9 =$ identité on F too. If F has an extremity x , then $\varphi(x)=x$, and again let P^* be F minus x and the edge abutting at x . Then F^* has all the properties P has, so $\varphi =$ identité on P^* , hence $\varphi =$ identité on P . If F has a vertex x on which only 2 edges abut, we have one of the two cases:

vertex x on which only 2 edges abut, we have one of the two cases:



$n_1 =$ number of edges reversed by φ .

Then, unless $\Gamma = \emptyset$, the Lefschetz fixed point formula reads:

$$n_0 + n_1 = b_0 - b_1 < 0,$$

which is impossible. Q.E.D.

(2) $\text{---} \text{---} \text{---} e_1 e_2$

In the first case, $\varphi(y)=y$, and let F^* be F minus x , e_1 and e_2 . Then $\varphi =$ identité on F^* and $\varphi(e_i) = e_i$ too, since if φ reverses the e_i 's, this contradicts (iii). In the second case, let F^* be P minus x , and with e_1 and e_2 identified: e . Then $\varphi =$ identité on F^* , so $\varphi =$ identité on P . Next, say F has an edge e with extremities x and y such that $\varphi(x)=x$, $\varphi(y)=y$, $\varphi(e)=e$. Soit P^* be P minus e . Then $\varphi =$ identité on F^* , so $\varphi =$ identité on P . If none of these reductions are possible, we must be in a situation where a) every vertex is the abutment of at least three edges and b) no edge is left fixed. It is easily seen that the first Betti number b_1 of any of the connexes composantes of P is at least 2. Soit $n_0 =$ number of fixed vertices $n_1 =$ number of edges reversed by $\sim$. Then, unless $F = \emptyset$, the Lefschetz fixed point formula reads: $n_0 + n_1 = b_0 - b_1 < 0$, which is impossible. Q.E.D.

We consider the situation: $K =$ discretely-valued corps; $1 \sim =$ integers in K , $k = 1 \sim / \sim J1 =$ residue corps (assumed algébriquement clos) ; $S = \text{Spec}(R)$, $\sim$ and s its generic and point fermés respectively; $C =$ a curve, lisse, géométriquement irréductible and propre over K , of genus $g \sim 2$; $J =$ the jacobian variety of C ; $og =$ the Néron model of J over R (cf. [N]); $J \sim$ the open subgroup schéma with $J \sim$ component of J_s ; $c \sim =$ the minimal model of C over R . A word about the existence and uniqueness of $c \sim$ is needed. We recall that $c \sim$ is to soit un régulier schéma, plat and propre over R , with fibre générique $c \sim_n = C$ such that t for any other régulier schéma $c \sim$, plat over P , with fibre générique $\sim f_n = C$, the birational map $c \sim, \sim c \sim$ is a morphisme, Šafarevich in [S] and Lichtenbaum [L] have proven that such a $\sim$ (which is obviously unique) exists, provided that there is some régulier $5'$, propre and plat over R , with fibre générique C . And, in fact, that $c \sim$ is projective over R . To construct such a $c \sim$, proceed as follows: first let $c \sim$, soit un régulier schéma, projective and plat over R with fibre générique C . Soit R be the completion of R , and let $c \sim, \sim c \sim, \times \text{spooRSpec } R$. Then O' is an excellent surface, so by [Ab] and by unpu-

published results of Hironaka, there is a sheaf of ideals $\mathcal{I}$ with support in the singular locus of O' such that blowing up $\mathcal{I}$ leads to a regular surface O . But then for some n , so $\sim$ is induced by a unique Sheaf of ideals $\mathcal{I}_0 \in e_n$. So $\sim'$ be obtained by blowing up $\mathcal{I}$. Then $\sim, \sim' \in \text{cg}, \times \text{spe}, \times \text{Spec } R$. so cg is regular, $c\sim$ is also projective over R . hence a projective cg exists.

Définition (2. x). — J has stable reduction if J8 has no radical unipotent.

Définition (2.2). — C has stable reduction in sense i if $c \sim s$ is réduite and has only

points doubles ordinaires. C has stable reduction in sense 2 if there is a stable curve $\tilde{C}$, over R with fibre générique c_{gs} . Note that if a stable $\tilde{C}$ exists, then by Théorème (I. I) it is unique.

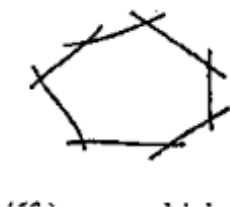
Proposition (2.3). — The two senses of stable reduction for C are equivalent.

Démonstration. — Say a stable cg, exists. Blowing up the singularities of $c_{\sim}$, as in

Lemme (i. 12), we obtain the minimal model $\tilde{C}$ of C and it is seen that $c_{\tilde{C}} \cdot 8$ is réductes

with only points doubles ordinaires. Conversely, suppose the minimal model $\tilde{\sim}$ has this property. Soit $E_1, \dots, E_n$ be the non-singular rational components of $\tilde{\sim}$ which meet the other components in only two points. Then the E_i 's divide into several chains of the type:

ed of E_i 's and has the type:



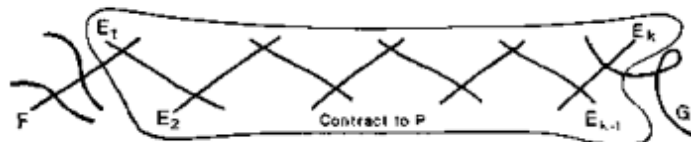
$$\begin{aligned} n &\geq 2 \\ \mathcal{C}_i &= \text{loop of } E_i\text{'s} \\ (E_i^2)_{\text{on } \mathcal{C}} &= -2. \end{aligned}$$

unless the entire fibre consisted of E_i 's and has the type: $n > 2 \sim 8 = \text{loop of } E_i\text{'s } (E\varphi)_{\text{oo}} = -2$. But in this case $\text{genus}(C) = \text{genus}(C_{\text{gs}}) = 1$, which contradicts our assumption. Now, according to Théorème (27. I) of Lipman [Li] (generalizing a result of Artin [A2], which works for surfaces of fini type over a corps) any set of k non-singular rational curves connexes in a chain

as above on a regular surface X , with self-intersection 2 on X , can be blown down to a rational double point P of type A_k on a normal surface X_0 . Reversing the process and blowing up a rational double point of type A_k , it is easy to see that nonsingular branches y on X , crossing transversally only the first or the last rational curve in the chain: $E_2 E_k \dots$

are still non-singular branches when mapped to X_0 ; and if Y_1, Y_2 intersect E_1 or E_k in distinct points, then they cross transversally on X_0 . Par conséquent, suppose we blow down all the chains of E_i 's on $c_{\sim}$. Soit $c_{\sim}$, be the normal surface so obtained. Then if one of these chains fits into $\sim$, like this:

fits into $\sim$, like this:



then on $\sim$, the images of F and G still have only points doubles ordinaires, each has one non-singular branch through the singular point P , and these branches cross transversally. Par conséquent $\sim'$ is a stable curve over R . Q.E.D. We are now ready to prove the key result on which our proof of irreducibility depends:

Théorème (2.4). — J has stable reduction if and only if C has stable reduction.

Démonstration. — The connection between $\sim$ and of is based on the following result of Raynaud [R] :

Théorème (2.5). — If $c_{\sim}$ and of are as above, and the greatest common denominator d of

the multiplicities of the composantes of $\sim$ is 1, then of_0 represents the foncteur $Pic_{\sim}$ (This result is not stated as such in [JR]. It comes out like this, in the terminology of this paper : a) Condition (N) is verified and $p_*(0_{\sim}) = 0_S$, so b) $\sim$ is cohomologically plat over S in $\dim. 0$ by Théorème 4; c) therefore $Pic_{\sim}$ is representable and séparé over S by Théorème 3; d) since $E = \{o\}$ and $Q = P/E$, we find $P_{\sim}$ and since $C_{\sim}$ is 1-dimensional over S , p_0 and Q_0 are lisse over S ; therefore $R_{\sim} = Q_0$ and $P_{\sim} = R$. e) Then by Théorème 5, $Pic_{\sim}$ is the identité component of the N -dron model of $Pic_{\sim} = J$.) Now assume that C has stable reduction. Then $\sim$ is réduite so $d=1$. By

Théorème (2.5), $of_{\sim 176}$ Par conséquent $of_{\sim 176}$). Puisque ~ 8 is réduites

with only points doubles ordinaires, its generalized jacobian $Pic_{\sim}$ is an extension of an abelian variety by a torus, i.e., has no radical unipotent. Par conséquent J has stable reduction. The converse is more difficult. Assume J has stable reduction. We first prove that C has stable reduction under the additional hypothesis that C has a K -rational point (1). In this case, $\sim$ has an R -rational point, and since $\sim$ is regular, sections of of over R pass through composantes of $\sim$ of multiplicity one. Par conséquent $d=1$, and (1) This is in fact the only case which will be needed in our application to questions of irreducibility.

9 ~ P. DELIGNE AND D. MUMFORD

Théorème (2.5) applies. In particular, $Pic_{\sim}$ is so $Pic_{\sim}$) has no unipotent

radical. We apply:

Lemme (2.6). — Soit D soit un complete i -dimensional schéma over k such that $H^0(\mathcal{O}_D) = k$,

and such that the generalized jacobian of D has no radical unipotent. Then (i) $\pi_0(\mathcal{O}_D) = X(\mathcal{O}_D^{\text{red}})$; (ii) the singularities of D_{red} are all transversal crossings of a set of non-singular branches (i.e., analytically isomorphe to the union of the coordinate axes in $\mathbb{A}^m$). Démonstration. — Soit $J \subset \mathcal{O}_D$ be the ideal of nilpotent elements. Filtering J by a chain of ideals J_k such that $J_k \subset J_{k+1}$, and using the exact sequence: $0 \rightarrow J_k/J_{k+1} \rightarrow \mathcal{O}_D/J_{k+1} \rightarrow \mathcal{O}_D/J_k \rightarrow 0$ it is easy pour déduire that $\text{Pic}_{\sim}$ is an extension of $\text{Pic}_{\sim}$ by a unipotent group. Par conséquent, since by assumption $\text{Pic}_{\sim}$ has no unipotent subgroups, $\text{Pic}_{\sim} = \text{Pic}_{\sim}$. Puisque $H^0(\mathcal{O}_D)$, resp. $H^0(\mathcal{O}_D^{\text{red}})$, is naturally isomorphe to the Zariski tangent space to $\text{Pic}_{\sim}$ resp. $\text{Pic}_{\sim}$ it follows that $H^0(\mathcal{O}_D) \cong H^0(\mathcal{O}_D^{\text{red}})$ hence (i) is proven. Soit $\pi : C \rightarrow D$ be the normalisation of D , and let D^* be the local ringed space which, as topological space is D , and whose structure sheaf is given by: $\mathcal{F}(U, \mathcal{O}_D) = \{f \in \mathcal{F}(U, \mathcal{O}_C) \mid x \sim x', f(x) = f(x') \text{ if } x \sim x'\}$. It is easy to check that D^* is a 1-dimensional schéma whose singularities are all transversal crossings of a set of non-singular branches and that π factors: $C \rightarrow D^* \rightarrow D^{\text{red}}$. I claim that $\pi' : D^* \rightarrow D^{\text{red}}$ is an isomorphism. Filter $\mathcal{O}_D/\mathcal{O}_D^{\text{red}}$ as to obtain a chain of coherent $\mathcal{O}_D$ -algebras: $(\mathcal{O}_D^{\text{red}} \subset \mathcal{O}_D^{\text{red}}(1) \subset \dots \subset \mathcal{O}_D^{\text{red}}(k) \subset \dots \subset \mathcal{O}_D^{\text{red}})$ such that $\pi'(\mathcal{O}_D^{\text{red}}(k)) = \mathcal{O}_D^{\text{red}}(k)$. Equivalently, this factors $\pi' : D^* \rightarrow D^{\text{red}}$ as $D^* \rightarrow D^{\text{red}}(1) \rightarrow \dots \rightarrow D^{\text{red}}(k) \rightarrow D^{\text{red}}$ where $\mathcal{O}_D^{\text{red}}(h) \subset \mathcal{O}_D^{\text{red}}(h+1)$, and all arrows are homeomorphisms. If $\{x\} = \text{Supp}(\mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}})$, then we get an exact sequence: $\mathcal{O}_D^{\text{red}} \rightarrow \mathcal{O}_D^{\text{red}}(1) \rightarrow \mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}} \rightarrow \mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}}(1) \rightarrow 0$ ($\mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}}$ denotes the residue corps at x , as sheaf on D), and hence: $1 \otimes \mathcal{O}_D^{\text{red}}(1) \rightarrow \mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}} \rightarrow \mathcal{O}_D^{\text{red}}(1)/\mathcal{O}_D^{\text{red}} \rightarrow 0$. It follows easily that $\text{Pic}_{\sim}$ is an extension of $\text{Pic}_{\sim}$ by $G_{\sim}$. But $\text{Pic}_{\sim}$ has no unipotent subgroups, and this can only happen if $D_{\sim}^{\text{red}} = D^*$. Q.E.D. for lemma.

We apply the lemma to $c_{\sim}$. According to Lichtenbaum [L] and Šafarevich [S], there is a divisor K on $\sim$ such that for all positive divisors D lying over the point fermé of $\text{Spec}(R)$, we have: $(D + K) \cdot z(eD) = 0$.

Soit $E_1, \dots, E_n$ be the composantes of $c_{\sim}$, $d_1, \dots, d_n$ their multiplicities. Then conclusion (i) of the lemma implies that: But $((-d_i E_i) \cdot E_k) = 0$, all k , since $-d_i E_i$ is the divisor of a function $r : \sim \rightarrow R$ if $(=) = \text{maximal ideal of } R$. Par conséquent : $((-d_i E_i) \cdot K) = (-E_i \cdot E_i)$. Note that at least one d_i equals i since $c_{\sim}$ has a section over $\text{Spec}(R)$ and every section must pass through a component of $c_{\sim}$ of multiplicity i . Moreover the intersection matrix $(E_i \cdot E_j)$ is negative indefinite, with one-dimensional degenerate subspace generated by $\sum d_i E_i$, hence $\sim$ some $d_i > i$, it follows that $(-E_i \cdot E_i) < 0$. Par conséquent, by (*), $(E_i \cdot K) < 0$ for some i . Then we have: a) $(E \cdot K) < 0$; b) $(E_i \cdot E_{i+1}) < 0$; c) hence in fact $(E \cdot E) = (E \cdot K) = 0$ so E is an exceptional curve of the first kind. This contradicts our assumption that on $\sim$ all possible curves have been blown down. Par conséquent $d_i = i$, all i . This proves that $\sim$ is réduite. By conclusion (ii) of the lemma, plus the fact that the dimension of its Zariski tangent-space is everywhere one or two (since $c_{\sim}$ lies on a regular surface $\sim$), we deduce that $c_{\sim}$ has only points doubles. This proves that C has stable reduction in the case that $C(K) \neq \emptyset$. In the general case, C will acquire a rational point in a fini extension K' of K . Soit S' be the spectrum of the localisation at some maximal ideal of the clôture intégrale of R in K' ; S' is the spectrum of a valuation ring and is faithfully flat over S . We put $S'' = S'$.

is S' and denote by C' and C'' the images respectively.

$$\begin{array}{ccccc} C'' & \xrightarrow{\quad} & C' & \xrightarrow{\quad} & C \\ \downarrow & & \downarrow & & \downarrow \\ S'' & \xrightarrow[p\pi_2]{p\pi_1} & S' & \xrightarrow{p} & S \end{array}$$

$\times$ and denote by C' and C'' the image inverses of C on $S' = \text{Spec}(K')$ and S respectively. $C'' \sim C' \text{ prl } S'' \sim S' \text{ p}\pi$; C is $> S$

Soit C' be the stable curve on S' having C' as fibre générique. The restriction, C' , of C' ! to S' carries a descent datum with respect to p , i.e. an isomorphism, φ , between the restrictions of $p\pi(C')$ and $p\pi(C')$ to S' . It remains only to extend the isomorphism φ to an isomorphism, φ , between the $p\pi(C')$. Then φ will soit un descent datum for C' with respect to p . As C' is canonically polarized (I.2), this descent datum will be effective, and so define a stable curve, C , over S with fibre générique C . Because j_0 has no radical unipotent, the image inverse, p_{j_0} , of j_0 on S' is the identité component of the Néron Model of the jacobian of C' , so that, defining $q = p \circ p_1 = p \circ p_2$, one has $\text{Pic}_0(C/S, \mathbb{Q}) \cong \text{Pic}_0(p\pi, \mathbb{Q}) \cong \text{Pic}_0(p_2 C) \cong q_{j_0}$. We denote by T the closed subschéma of $\text{Isom}(S'', p\pi(C'), p\pi(C'))$ corresponding to those isomorphismes which induce (via the preceding identifications) the identité on the image inverse of j_0 . By (I.11), T is fini and unramified over S'' , and by (I.13) T is radical over S'' . We conclude that the morphisme from T to S'' identifies T with a closed subschéma X of S'' . As X contains S , which is schematically dense in S , we have that X is S'' and T is the desired section, η , of $\text{Isom}(S'', \text{pr}_1^* p\pi(C'), \text{pr}_2^* p\pi(C'))$ over S'' which extends $\sim \varphi$. Q.E.D. Combining Proposition (2.3), Théorème (2.4), and the théorème de réduction stable for variétés abéliennes quoted in the introduction, we obtain the most important consequence:

Corollaire (2.7). — Soit R soit un anneau de valuation discrète with quotient fmd K . Soit C soit un

lisse geometrically irréductible curve over K of genus $g > 2$. Then there exists a extension algébrique finie L of K and a stable curve WL over RL , the clôture intégrale of R in L , with generic fibre $CC_L \sim C \otimes K_L$. § 3" Elementary derivation of the theorem. Soit k soit un algébriquement clos corps of char. $p \neq 0$. We use the notation of w I, except that we will now denote by H_g the product $H_g \times$ of the previous H_g with $\text{Spec}(k)$: it is a disjoint union of non-singular varieties $H_{g,1}, \dots, H_{g,n}$ over k and $\eta_{g,0}$ is the subschéma of $H_{g,0}/k$ of tri-canonical stable curves. Similarly, H_a is the open dense subset of H_g of tri-canonical non-courbes singulières. By the results of [M1], we know that a coarse geometric quotient $\text{Hom}/\text{PGL}(5g-6) \rightarrow M_g$ exists, that it is a disjoint union of normal varieties over k and is the coarse espace de modules $\eta_{g,0}$ for non-courbes singulières of genus g . Soit $S = H_g - H_{g,0}$. Then everything decomposes into the same set of composantes.

Soit $H_{g,i}$ composantes of $H_g - H_{g,0}$ then $H_{g,i} \rightarrow H_{g,i}$ ca 0 and $0 \rightarrow M_{g,i} \rightarrow H_{g,i}/\text{dPGL}(5g-6) \rightarrow$ composantes of M_g and S^* disjoint union of $S_1, \dots, S_n, S_i \rightarrow H_{g,i}$ ica S . We want to prove that M_g or equivalently H_g or equivalently H_g is irréductible. We

shall use: (i) the fact that these statements are true in char. 0; (ii) the inductive assumption that these statements are true for smaller genus. Step I. — No component of $M \sim$ is complete (i.e., propre over k). Démonstration. — Here we use the char. 0 result. By [M1], there is a schéma X , quasiprojective over $\text{Spec}(W(k))$, $W(k)$ the Witt vectors, whose fibre fermée is $M \sim$ and whose fibre générique $X \sim$ is the char. 0 coarse espace de modules over the corps des fractions of $W(k)$. In particular, X , is known to be connexes. Puisque X is quasi-projective over $W(k)$, we can embed X as an open dense subset of a schéma X projective over $W(k)$. $X \sim$ is still connexes, hence by the théorème de connexité of Enriques-Zariski [EGA 3], the fibre fermée ${}^{\sim}0$ of ${}^{\sim}0$ is connexes. But if Y were a complete variety which is a component of M_g then: a) Y is an ouvert of M_g , which is an ouvert of ${}^{\sim}0$, and: b) since Y is propre/ k , Y would soit un closed subset of ${}^{\sim}0$ too. Par conséquent, $Y = {}^{\sim}X0$, hence M_g is itself irréductible and complete. On the other hand, if Ag is the coarse espace de modules of principally polarized g -dimensional variétés abéliennes, then the map associating to each curve its jacobian defines a morphisme: $0 : M_g \rightarrow Ag$. If M_g were complete, the image of 0 would be closed. But it is well known that the closure of the image of 0 contains all products of lower dimensional jacobians too, so it is not closed. Q.E.D. Step II. — No component of H_g consists entirely of non-courbes singulières, i.e., $S \sim = O$ for all i . Démonstration. — Here we combine Step I with the result of § 2. Take any i . Soit $T = \text{Spec } k[[t]]$. Puisque $M \sim$ is not complete, there is a morphisme φ of the generic point T , of T into M_g which does not extend to a morphisme of T into M_g . Now we replace $T \ni 0$ by its normalisation T' in a extension algébrique finie and let $\varphi' : T \rightarrow M_g$ be the

induced morphisme, which still does not extend to a morphisme from T' to M_g . By the results of w 2, if T' is chosen suitably there exists a stable curve $\gamma : C' \rightarrow T'$ over T' whose fibre générique C , is a non-singular curve corresponding to the morphisme $r : T \rightarrow M_g$ via the foncteurial propriétés of the espace de modules. Puisque T' is the spectrum of a local ring, we can choose an isomorphisme $P : (\pi^*(\omega_{C'/T'}) \otimes P \otimes g^*G) \otimes P \otimes g^*G \times$ and get a tri-canonical embedding $C' \rightarrow \text{CPSg} \times \dots \times C'$, with this embedding, is then

induced from the universal tri-canonically embedded stable curve by a morphisme $t : T' \rightarrow H_g$. Puisque the fibre générique of C' is C , we get a commutative diagram:

THE IRREDUCIBILITY OF THE SPACE OF CURVES OF GIVEN

Let

$H_{g,i}$ = components of H_g

$H^0_i = H_{g,i} \cap H^0 =$ components of H^0

+ $T' \rightarrow H_g \cup \dots \cup \pi^* S$, $0 \sim H_g \rightarrow M \sim C$: $o \sim 9 \sim M9$ If $\text{Im}(qJ) \subset H \sim$ then φ' would extend to a morphisme from T' to $M \sim$ hence from T' to M_g and this is a contradiction. Moreover, $\sim(T_i)$ must soit un point of H^0 since its image in $M \sim$ is in $M \sim$. Par conséquent the image x of the point fermé of T' by $\sim$ is in the closure

of $H_{g,i}$, i.e., in $H_{g,i}$, but not in $H^0_{g,i}$ itself. Q.E.D. Step III — S^* is connexes. Démonstration. — This will follow using only the induction assumption of irreducibility for lower genera. Soit $Z \subset H_g \times 5g-G$ be the universal tri-canonically embedded stable curve. Soit ScZ be the set of points where Z is not lisse over H_g . As we proved in w I, S is non-singular and is the normalisation of S^* . In particular, this shows that if $x \in S^*$, then the corresponding

curve Zx has exactly one double point if and only if x is a non-singular point of S^* . Stable curves C of genus g with exactly one double point belong to one of the following types:

generic fibre of C' is C'_η , we get a

$$\begin{array}{ccc} T' & \xrightarrow{\psi} & H_g \\ U & & U \\ T'_i & \xrightarrow{\tau \circ \psi|_i} & H_g^0 \\ & \searrow \varphi' & \downarrow \\ & & M_{g,i}^0 \subset M_g^0 \end{array}$$

φ' extend to a morphism from T' to

type 0: $C \sim$ type k : C irréductible normalisation C' of C has genus $g-1$. C has two non-singular composantes C_1, C_2 $\text{genus}(C_1) = k$ $\text{genus}(C_2) = g-k$ if one double point is of type 0. A non open dense subset of S^* of non-singular points is the disjoint union of ouverts $S^*(o), \dots, S^*(l/g/)$. We first check: (.) Each set $S^*(k)$ is irréductible. Démonstration of (.)- Take the case $k=0$: the cases $1 \leq k < [g]$ are similar. Soit $T = \{(x_1, x_2) \mid 1+x_2 \text{ and } \pi(x_1) = \pi(x_2) \in H^0_{g-1}\} \subset Zg_{g-1} \times H^0_{g-1}$. T is lisse with irréductible fibres over H^0_{g-1} hence T is irréductible. Consider the correspondence relating $S^*(o)$ and T :

, set of pairs

type 0:		C irréductible normalization C' of C has genus $g-1$.
type k :		C has two non-singular components C_1, C_2 $\text{genus}(C_1) = k$ $\text{genus}(C_2) = g-k$
$1 \leq k \leq \left\lfloor \frac{g}{2} \right\rfloor$		

$x \in S^*(o)$, $\{x_1, x_2\} \in T$ such that if $y = \pi(x_1) = \pi(x_2)$, $j \in I$ then there exists a birational morphisme $\sim$ such that $f(x_1) = f(x_2)$. It is easy to check that W is Zariski-closed. Moreover, for any $\{x_1, x_2\} \in T$, $W \cap (S^*(o) \times \{x_1, x_2\})$ is an orbit in $S^*(o)$ under $\text{PGL}(5g-6)$: so these are non-empty irréductible subsets all of the same dimension. It follows that W itself is irréductible. But the projection from W to $S^*(o)$ is surjectif, so $S^*(o)$ is irréductible. Q.E.D. for (.). Now for any k , $1 \leq k < [g/2]$, choose any stable curve $C(k)$ of the type:

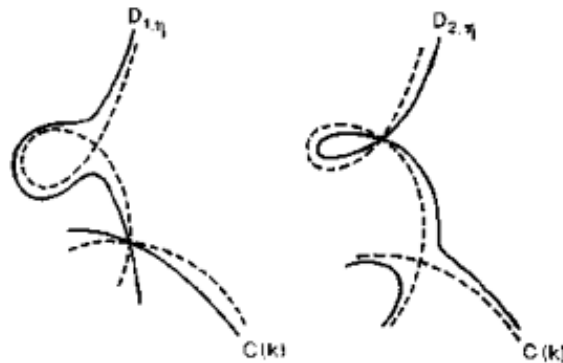
$$W = \left\{ \begin{array}{l} \text{set of pairs } x \in S^*(o), \{x_1, x_2\} \in T \text{ such that if } y = \pi(x_1) = \pi(x_2), \\ \text{then there exists a birational morphism} \\ f: (Z_{y-1})_y \rightarrow (Z_y)_x \\ \text{such that } f(x_1) = f(x_2). \end{array} \right\}$$

it is easy to check that W is Zariski-closed. Moreover for any $\{x, x'\}$

On Q , one non-singular component $C(k)'$ of genus k . On Q_2 one component $C(k)''$ with one double point, normalisation of genus $g-k-1$. Soit $P(k)$ soit un point of H_g such that $Z_p(k) \sim C(k)$. Step III will be completed if we prove : (**). $P(k)$ is in the closure of $S^*(o)$ and of $S^*(k)$, hence $S^*(o), S^*(k)$ both lie in the same topological component of S^* . Démonstration of (**). — Soit $T = \text{Spec } k[[t]]$. Using the fact that S^* has two branches through $P(k)$, one for each of the points doubles of $C(k)$, we see that there exist two morphismes f_1, f_2

for each of the double points of $C(k)$, we see that

$$f_1, f_2 : T \rightarrow S^* \\ f_1(T_s) = f_2(T_s) = P(k), \quad T_s = \text{closed point of } T$$



$f_1 : T \rightarrow S^*$, $f_1(T_s) = f_2(T_s) = P(k)$, $T_s = \text{point fermé of } T$. D_1, D_2 are the two stable curves over T induced by f_1 and f_2 , then: a) the fibre fermées D_1, D_2 are $C(k)$; b) there are sections $s_1 : T \rightarrow D_1, s_2 : T \rightarrow D_2$ whose images are non-lisse points of rel and such that $s_1(T_s)$ and $s_2(T_s)$ are the two points doubles Q_1 and Q_2 of $C(k)$ respectively, and: c) the fibre génériques D_1, D_2 have only one double point (cf. figure). I claim: A) D_1 is of type (k); B) D_2 is of type (0).

To prove A), let D_1' be the result of blowing up the subschéma $s_1(T)$ of D_1 . Then D_1' is still plat and propre over T and its fibre spéciale is the fibre spéciale of D_1 with Q_1 blown up (use the fact that formally at Q_1 , D_1 is isomorphe to $k[[t, x, y]]/(x, y)$ with the section given by $x=y=0$). Par conséquent the fibre spéciale of D_1' is the disjoint union of $C(k)', C(k)''$, so the general fibre of D_1' is the disjoint union of two irréductible curves which specialize to $C(k)', C(k)''$ respectively. Puisque D_1 has only one double point, D_1' is non-singular, so D_1' is the disjoint union of two non-singular irréductible curves which must then have the same genera as $C(k)'$ and $C(k)''$, i.e., $k, g-k$. Thus D_1 has type (k). To prove B), it suffices to check that D_2 is geometrically irréductible. If not, D_2 would have two composantes meeting at the single point $s_2(T_n)$. Puisque D_2 is lisse over T at each generic point of its fibre spéciale, distinct

geometric composantes of D_2 , have to have specializations which are distinct composantes of $(D_2)_s$. Then $(D_2)_s$ would have two composantes meeting at the point $Q_2 = s_2(T_s)$. This is false, so B) is proven. Now because of A), $\text{fl}(T_s) \in S^*(k)$, hence $P(k) = \text{fl}(T_s)$ is in the closure of $S^*(k)$. And because of B), $\text{f}_2(T_s) \in S^*(o)$, hence $P(k) = \text{f}_2(T_s)$ is in the closure of $S^*(o)$ too. Q.E.D. for (**). This completes the proof of Step III since we now see that all irréductible composantes of S^* are part of the same topological component. Finally, from Steps II and III, we see that a) S^* , being connexes, is part of a single component of H_g , while b) each component of H_g contains part of S^* . Thus H_g is irréductible, as was to be proven. § 4"

(ii) a section η of $\mathcal{S}_0$ over (U, φ) .

Definition (4.2). — A 1-morphism of stacks over C , $F : \mathcal{S}_1 \rightarrow \mathcal{S}_2$, will be called representable if for any X in C and any 1-morphism $x : X \rightarrow \mathcal{S}_2$, the fibre product $X \times_{\mathcal{S}_2} \mathcal{S}_1$ is a representable stack.

In down to earth terms, this means the following:

- (i) for any $f : Y \rightarrow X$ in C , the category whose objects are pairs
 $\{ \text{a section } y, \text{ of } \mathcal{S}_1 \text{ over } Y; \text{ an isomorphism } F(y) \xrightarrow{\sim} f^*(x) \}$

is equivalent to a category $S(f)$ in which all morphisms are identities;

is equivalent to a category $S(f)$ in which all morphisms are identities;

- (ii) the functor $f \mapsto \text{Ob } S(f)$ is representable by some $g : Z \rightarrow X$. Such a Z represents the fibre product $X \times_{\mathcal{S}_2} \mathcal{S}_1$.

Let P be a property of morphisms in C , stable by change of base and of a local nature on the target.

Definition (4.3). — A representable morphism $F : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ of stacks over C has property P if for any 1-morphism $x : X \rightarrow \mathcal{S}_2$ the morphism in C deduced by base change: $F' : X \times_{\mathcal{S}_2} \mathcal{S}_1 \rightarrow X$ has that property.

Proposition (4.4). — Let $\mathcal{S}$ be a stack. The diagonal map

$$\mathcal{S} \rightarrow \mathcal{S} \times \mathcal{S}$$

is representable if and only if for all $X, Y \in \text{Ob } C$ and 1-morphisms $x : X \rightarrow \mathcal{S}$ $y : Y \rightarrow \mathcal{S}$

pology. This means that, f

$$\begin{array}{ccc} X_i & \xrightarrow{g_i} & X \\ \downarrow f_i & & \downarrow f \\ Y_i & \xrightarrow{h_i} & Y \end{array}$$

e and cover X (resp. Y):

where the g_i (resp. h_i) are etale and cover X (resp. Y):

$$P(f) \Leftrightarrow \forall_i P(f_i).$$

y, commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & T_1 \\ f' \downarrow & & \downarrow f \\ Y & \xrightarrow{y} & T_2 \end{array}$$

and x, y are etale and surjective

separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a “proper morphism” and the analogue of Chow’s lemma.

A morphism $f: T_1 \rightarrow T_2$ is said to have a property P , locally on T_2 , if there exists a surjective etale morphism $x: X \rightarrow T_2$ such that the morphism f' deduced from f by the change of base by x has property P .

Definition (4.11). — A morphism $f: T_1 \rightarrow T_2$ is proper if it is separated, of finite type and if, locally over T_2 , there exists commutative diagrams

$$\begin{array}{ccc} T_3 & \xrightarrow{g} & T_1 \\ h \searrow & & \swarrow f \\ & T_2 & \end{array}$$

with g surjective and h representable and proper.

with g surjective and h representable and proper.

The following form of Chow’s Lemma will be sufficient for our purposes.

Theorem (4.12). — Let S be a noetherian scheme and f be a morphism from an etale site T to S . We assume f to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{array}{ccccc} T & \xleftarrow{\theta} & T' & \xrightarrow{j} & T'' \\ \downarrow f & \nearrow f' & & \nearrow f'' & \\ S & & & & \end{array}$$

in which T' and T'' are schemes and such that

where θ is surjective and θ_1, θ_2 are isomorphisms

$$\begin{array}{ccc} & & T \\ \theta_1 \nearrow & & \downarrow f \\ \text{Spec}(V) & \longrightarrow & S \end{array}$$

where θ_1 and θ_2 are isomorphisms

Définition (4. I). — Soit C soit un catégorie with a Grothendieck topology. We assume products

and fibre products exist in C . A champ in groupoids over C is a catégorie over C , $p : \mathcal{F} \rightarrow C$ such that: (i) $\mathcal{F}$ is fibered in groupoids over C . (ii) For any $U \in \text{Ob } C$ and any objets x, y in $\mathcal{F}_U$ the foncteur from C/U to (sets) which to any $\varphi : V \rightarrow U$ associates $H^0_{\text{ms}}(V, \varphi^*x, \varphi^*y)$ is a sheaf. (iii) If $\varphi : V \rightarrow U$ is a covering family in C , any descent datum relative to the φ_i , for objets in $\mathcal{F}$, is effective. For each $x \in \text{Ob } \mathcal{F}$ there are given isomorphismes between the image inverses of $x_i = \varphi_i^*x$ and $x_j = \varphi_j^*x$ over $V_0 = V_i \times_C V_j$ and the pull-backs of these isomorphismes on $V_i/k = V_i \times_C V_j$ satisfy a " ω -cycle" condition. In (iii) it is required that reciprocally, any such "descent datum" be defined by some $x \in S$. In whatfollows, for the sake of brevity, we will use "champ" to mean "champ in groupoids". If $U \in \text{Ob } C$ and if $\mathcal{F} \rightarrow C$ is a champ over C , the fibre $\mathcal{F}_U$ will be called the catégorie of sections of $\mathcal{F}$ over U . Soit C soit uns in (4. I)- The champs over C are the objets of a 2-catégorie [B] (champs/ C): i-morphismes are foncteurs from one champ to another, compatible with the projection into C ; and 2-morphismes are morphismes of foncteurs. In this 2-catégorie every 2-morphisme is an isomorphism. Products and 2-fibre products exist in this 2-catégorie. To each $X \in \text{Ob } C$ is associated the "representable" champ over C whose catégorie of sections over U is the discrete catégorie whose objets are the morphismes from U to X . This champ will be denoted simply X . For any champ $\mathcal{F}$, the catégorie $H^0_{\text{m}}(X, \mathcal{F})$ is canonically equivalent to the catégorie of sections of $\mathcal{F} \times_C X$ over X . Because of this, $\mathcal{F}$ is sometimes said to "classify" its sections over variable $X \in \text{Ob } C$. In the catégorie $H^0_{\text{m}}(\mathcal{F}, X)$, all morphismes are identities, i.e., $H^0_{\text{rn}}(\mathcal{F}, X)$ is just a set. Soit us denote also by C the 2-catégorie having the same objets and morphismes as C , and in which the identities are the only 2-morphismes. The above construction then identifies C with a full sub-2-catégorie of (champs/ C).

For each $S \in \text{Ob } C$, the catégorie C/S satisfies the assumptions of (4.I). Any champ $\mathcal{F}$ over C/S (an "S-champ") gives rise to a champ $\mathcal{F}_S$ over C ; a section $(\varphi, \sim)$ of $\mathcal{F}$ over $U \in \text{Ob } C$ consists of (i) a morphisme $\varphi : U \rightarrow S$; (ii) a section $\sim$ of $\mathcal{F}_U$ over (U, φ) .

Définition (4.2). — A I-morphisme of champs over C , $F : \mathcal{F}_1 \rightarrow \mathcal{F}_2$, will be called repre-

sentable if for any X in C and any i-morphisme $x : X \rightarrow \mathcal{F}_2$, the fibre product $X \times_{\mathcal{F}_2} \mathcal{F}_1$ is a representable champ. In down to earth terms, this means the following: (i) for any $f : Y \rightarrow X$ in C , the catégorie whose objets are pairs $\{ \text{a section, } y, \text{ of } \mathcal{F}_1 \text{ over } Y; \text{ an isomorphism } F(y) \sim f^*(x) \}$ is equivalent to a catégorie $S(f)$ in which all morphismes are identities; (ii) the foncteur $f \in \text{Ob } S(f)$ is representable by some $g : Z \rightarrow X$. Such a Z represents the fibre product $X \times_{\mathcal{F}_2} \mathcal{F}_1$. Soit/' soit un propriety of morphismes in C , stable by change of base and of a local nature on the target.

Définition (4.3). — A representable morphisme $F : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ of champs over C has propriety t

if for any I-morphisme $x : X \rightarrow \mathcal{F}_2$ the morphisme in C deduced by changement de base: $F' : X \times_{\mathcal{F}_2} \mathcal{F}_1 \rightarrow X$ has that propriety.

Proposition (4.4). — Soit $\mathcal{F}$ soit un champ. The diagonal map

$\mathcal{F} \rightarrow \mathcal{F}$. $\mathcal{F} \times \mathcal{F}$ is representable if and only if for all $X, Y \in \text{Ob } C$ and i-morphismes $x : X \rightarrow \mathcal{F}$, $y : Y \rightarrow \mathcal{F}$, the fibre product $X \times_{\mathcal{F}} Y$ is representable. If $X \in \text{Ob } C$ and x, y are sections of $\mathcal{F}$ over X , we denote by $\text{Isom}(X, x, y)$ the sheaf on C/X which to every Z over X associates the set of isomorphismes between the image inverses of x and y over Z . Then the objet representing $\mathcal{F} \times \mathcal{F}$ (the prodct taken with the map $(x, y) : X \rightarrow \mathcal{F} \times \mathcal{F}$ is just $\text{Isom}(X, x, y)$. If $x : X \rightarrow \mathcal{F}$, $y : Y \rightarrow \mathcal{F}$ are I-morphismes, then the objet representing $X \times_{\mathcal{F}} Y$ is just $\text{Isom}(X \times Y, x \times y)$. Henceforth, C will be the catégorie des schémas with the étale topology (SGAD, IV, (6.3)).

Définition (4.5). — A champ $\mathcal{S}$ is quasi-séparé /f the diagonal morphisme from to $\mathcal{S}$ to $\mathcal{S} \times \mathcal{S}$ is representable, quasi-compact and séparé.

Définition (4.6). — A champ $\mathcal{S}$ is an algebraic champ (1) /f

(i) $\mathcal{S} \rightarrow \mathcal{S} \times \mathcal{S}$ is representable; (ii) there exists a i -morphism $x : X \rightarrow \mathcal{S}$ such that for all $y : Y \rightarrow \mathcal{S}$, the projection morphisme $X \times \mathcal{S} \rightarrow Y \times \mathcal{S}$ is surjectif and étale (i.e., x is étale and surjectif). (1) This definition is the "right" one only for quasi-séparé champs. It will however be sufficient for our purposes.

The 2-catégorie of champs algébriques contains the representable champs and is stable under products and fibre products. If $\mathcal{S}$ is a quasi-séparé algebraic champ, the diagonal map is unramified and quasi-affine.

Définition (4.7). — An algebraic champ $\mathcal{S}$ is séparé /f, $\mathcal{S} \rightarrow \mathcal{S}$ is propre (or,

equivalently, fini). A i -morphism $f : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ is séparé (resp. quasi-séparé) if for any morphism $x : X \rightarrow \mathcal{S}_2$ from a séparé schéma X to $\mathcal{S}_2$, the fibre product $\mathcal{S}_1 \times_{\mathcal{S}_2} X$ is séparé (resp. quasi-séparé). Example (4.8). — Soit X soit un schéma over S . Soit G soit un group schéma over S , étale, séparé and of fini type over S , which operates on X . We will denote by $[X/G]$ the S -champ whose catégorie of sections over an S -schéma T is the catégorie of principal homogeneous spaces (p.h.s.) E over T , with structural group G (i.e., a p.h.s. under GT), provided with a G -morphism $\varphi : E \rightarrow X$. The principal homogeneous space $G \times X$ over X (G acting only on the first factor) plus the G -morphism $G \times X \rightarrow X$ (given by the action of G on X) is a section of $[X/G]$ over X . The corresponding morphisme $q : X \rightarrow [X/G]$ is étale and surjectif, so that $[X/G]$ is an algebraic champ. In addition, X is a principal homogeneous space over $[X/G]$; the champ $[X/G]$ is representable if and only if X is a principal homogeneous space over a schéma Y , in which case $[X/G] \cong Y$. If $X = S$, then $[X/G] = [S/G]$ might be called the "classifying champ" of G over S . Example (4.9). — Supposons que a champ $\mathcal{S}$ has the property that in each catégorie $\mathcal{S}_x$ the only morphismes are the identité morphismes. Then $2\mathcal{S}_x$ is just a set $\mathcal{O}_x(X) = \text{Ob } \mathcal{S}_x$ and this set, under pull-back, is a contravariant foncteur $\mathcal{O}_x$ in X . Conditions (ii) and (iii) of (4.1) assert that the foncteur $\mathcal{O}_x$ is a sheaf on C . Artin and Knutson [K] have defined an algebraic space to soit un sheaf $\mathcal{O}_x$ such that: (i) for any morphismes $X \rightarrow \mathcal{O}_x$, $Y \rightarrow \mathcal{O}_x$ of representable foncteurs to $\mathcal{O}_x$, the fibre product $X \times_{\mathcal{O}_x} Y$ is representable; (ii) there exists a morphisme $X \rightarrow \mathcal{O}_x$, represented by surjectif, étale morphismes of schémas. This is exactly what we have called an algebraic champ in this case.

Définition (4.10). — Soit $\mathcal{S}$ soit un algebraic champ. The étale site $\mathcal{S}_{\text{ét}}$ of $\mathcal{S}$ is

the catégorie with objets the étale morphismes $x : X \rightarrow \mathcal{S}$ and where a morphisme from (X, x) to (Y, y) is a morphisme of schémas $f : X \rightarrow Y$ plus a 2-morphisme between the i -morphism $x : X \rightarrow \mathcal{S}$ and $y \circ f : X \rightarrow \mathcal{S}$. A collection of morphismes $f_i : (X_i, x_i) \rightarrow (X, x)$ is a coveringfamily if the underlying family of morphismes of schémas is surjectif. The site $\mathcal{S}_{\text{ét}}$ is in a natural way ringed. When we speak of sheaves on $\mathcal{S}_{\text{ét}}$ we mean sheaves on Sect .

IOO P. DELIGNE AND D. MUMFORD We now explain how many concepts from la théorie des schémas may soit unplied to champs algébriques. Soit p soit un property of morphismes of schémas, stable by étale change of base, and of a local nature (for the étale topology) on the target. For instance: being an open immersion with dense image, being dominant, birational... A representable morphisme of champs algébriques $f : T_1 \rightarrow T_2$, is said to have property P if for one (and hence for every) surjectif étale morphisme $x : X \rightarrow T_2$, the morphisme of schémas deduced by changement de base $f' : X \times_{T_2} T_1 \rightarrow X$ has that property. Soit P soit un property of morphismes of

schémas, which, at source and target, is of a local nature for the étale topology. This means that, for any family of commutative squares $X_i \xrightarrow{g_i} X \xleftarrow{h_i} Y_i$ where the g_i (resp. h_i) are étale and cover X (resp. Y): $P(f) \Rightarrow \bigvee_i \text{Vie}(f/g_i)$. For instance: f plat, lisse, étale, unramified, normal, localement de fini type, localement de fini présentation. If $f: T_1 \rightarrow T_2$ is a morphisme of champs algébriques, we say f has property P if for one, then necessarily for every, commutative diagram $X \xrightarrow{x} T_1 \xleftarrow{y} Y \xrightarrow{u} T_2$ where X and Y are schémas and x, y are étale and surjectif, f has property P . Similarly, if P is a property of schémas, of a local nature for the étale topology, an algebraic champ T will be said to have property P if for one (and hence for every) surjectif étale morphisme $x: X \rightarrow T$, X has property P . This applies to, for instance, the properties of being regular, normal, localement noetherian, of characteristic p , réduites, Cohen-Macaulay... An algebraic champ T will be called quasi-compact if there exists a surjectif étale morphisme $x: X \rightarrow T$ with X quasi-compact. A morphisme $f: T_1 \rightarrow T_2$, of champs algébriques will be called quasi-compact if for any quasi-compact schéma X over T_2 , the fiber product $T_1 \times_{T_2} X$ is quasi-compact. It is enough to test the condition for a surjectif family $f_i: X_i \rightarrow T_2$. We define a morphisme $f: T_1 \rightarrow T_2$ to be of fini type, if it is quasicompact, and localement de fini type; of fini présentation, if it is quasi-compact, quasi-

Définition (4-II). — A morphisme $f: T_1 \rightarrow T_2$ is propre / f it is séparé, of fini

type and if, localement over T_2 , there exists commutative diagrams $T_3 \rightrightarrows T_1$ with g surjectif and h representable and propre. The following form of Chow's Lemme will be sufficient for our purposes.

Théorème (4-I2). — Soit S soit un noetherian schéma and f soit un morphisme from an étale site T

to S . We assume f to be séparé and of fini type. Then, there exists a commutative diagram $T \xrightarrow{g} T' \rightrightarrows T'' \xrightarrow{h} T$ in which T' and T'' are schémas and such that (i) g is propre, surjectif and generically fini; (ii) j is an open immersion; (iii) f'' is projective. Using (4. I2), it is easy to extend the cohomological theory of faisceaux cohérents to the present situation. In fact if $f: T_1 \rightarrow T_2$ is a propre morphisme of noetherian champs algébriques and if $\mathcal{O}$ is a coherent sheaf on T_2 , then the $R^i f_* (\mathcal{O})$ are faisceaux cohérents on T_2 . However, the $R^i f_* (\mathcal{O})$ don't need to be zero for i large enough. Soit S soit un schéma and G a fini group. We denote by p the projection $p: [S/G] \rightarrow S$. Quasi-coherent (resp. coherent) sheaves of modules on $[S/G]$ may (and will) be identified with quasicohérent (resp. cohérent) sheaves of modules on S , on which G acts. One has $R^i p_* (\mathcal{F}) \cong H^i(G, \mathcal{F})$. In general, the (quasi-coherent sheaf) cohomology of champs algébriques appears as a mixture of fini group cohomology and of schéma cohomology. The disjoint sum T of a family $(T_i)_{i \in I}$ of champs is the champ a section of which over a schéma X consists of (i) a decomposition $X = \bigsqcup_i X_i$; (ii) a section of T_i over X_i for each i .

102 P. DELIGNE AND D. MUMFORD The void champ $\emptyset$ is the one represented by the void schéma. A champ is connexes if it is non-void and is not the disjoint sum of two non-void champs.

Proposition (4. x3). — A localement noetherian algebraic champ is in one and only one way the

disjoint sum of a family of connexes champs algébriques (called its connexes composantes). We denote by $n_0(T)$ the set of connexes composantes of localement noetherian algebraic champ T . If $x: X \rightarrow T$ is surjectif and étale, $n_0(T)$ is the cokernel of the two maps $\sim_{\mathcal{O}}(X \times X) \rightarrow \sim_{\mathcal{O}}(X) \rightarrow \sim_{\mathcal{O}}(T)$.

Proposition (4. x4). — Soit T soit un algebraic champ of fini type over a corps k . Then,

T is connexes if and only if there exists a connexes schéma X , offini type over k , and a surjectif morphisme from X to T . An ouvert U of an algebraic champ T is a full subcatégorie $U \subset T$ which is an algebraic champ, which contains together with any $t \in \text{teOb}(T)$ all isomorphe t' and such that the inclusion $j: U \rightarrow T$ is representable by open immersions. The ouverts of T corresponds bijectively to the ouverts of its étale site. For each ouvert U of T , there exists one and only one full subcatégorie $T \dashv U$ of T , which is an algebraic champ, which contains together with any $t \in \text{teOb}(T)$ all isomorphe t' and such that (i) $T \dashv U$ is réduites; (ii) the inclusion map $i: T \dashv U \rightarrow T$ is representable by closed immersions; (iii) for any étale surjectif morphisme $x: X \rightarrow T$, the image inverse of $T \dashv U$ on X is the complement of the image inverse of U . An algebraic champ F in T satisfying (i) and (ii) is a closed subset of T , and the foncteur $U \sim T \dashv U$ is an isomorphisme of the set of open and the set of closed subsets of T . If F satisfies only (ii), $F \pi, \sim$ satisfies (i) and (ii) so that F defines a closed subset of T . An algebraic champ T is irréductible if it is not the union of two closed subsets, non void and distinct from T .

Proposition (4. x5). - - A noetherian algebraic champ T is in one and only one way the union

of irréductible closed subsets, none of which contains any other. They are called the irréductible composantes of T . If U is an open dense subset of T , the irréductible composantes of U are the non-void intersections of U with the irréductible composantes of T . Each irréductible component Of T is contained in a connexes component of T . Conversely:

Proposition (4. x6). — The connexes composantes of a normal noetherian algebraic champ

are irréductible.

Théorème (4. x7). I Soit f soit un morphisme of fini type from an algebraic champ T to a

noetherian schéma S . For $s \in S$, let $n(s)$ be the number of connexes composantes of the fibre géométrique of T at s . Then (i) $n(s)$ is a constructible function of s ;

Théorème (4. z8) (Valuative criterion for separation.) — The morphisme f is séparé

if and only if, for any complete anneau de valuation discrète with algébriquement clos residue corps and any commutative diagram $T \text{ Spec}(V) \sim S$ any isomorphisme between the restrictions of g_1 and g_2 to the generic point of $\text{Spec}(V)$ can be extended to an isomorphisme between g_1 and g_2 . This criterion is nothing other but the valuative criterion of propreness (EGA, II, 7.3 .8) applied to the (representable) diagonal morphisme.

Théorème (4. I9) (Valuative criterion for propreness.) — If f is séparé, then f is

propre if and only if, for any anneau de valuation discrète V with corps offractions K and any commutative diagram $\text{Spec}(K) \rightarrow \text{Spec}(V) \rightarrow S$ there exists a fini extension K' of K such that g extends to $\text{Spec}(V')$, where V' is the clôture intégrale of V in $K' // \text{Spec}(K') \rightarrow \text{Spec}(V)$. To prove a given f is propre, it suffices to verify the above criterion under the additional hypothesis that V is complete and has an algébriquement clos residue corps. Further, given a dense ouvert U of T , it is enough to test only g 's which factor through U .

Proposition (4.20). — Soit $\mathcal{F}$ soit un $\mathcal{O}_S$ -algèbre. Le foncteur $\mathcal{F}^*$ associe à tout $\mathcal{O}_S$ -module $\mathcal{M}$ le $\mathcal{F}^*\mathcal{M}$ et induit une équivalence de catégories entre : (i) la catégorie des champs algébriques représentables et affines sur $\mathcal{F}^*$; (ii) la catégorie des champs quasi-cohérents $\mathcal{O}_S$ -algèbres.

04 P. DELIGNE AND D. MUMFORD Soit $\mathcal{F}$ soit un $\mathcal{O}_S$ -algèbre. Pour chaque morphisme étale $x : U \rightarrow \mathcal{F}$ avec U affine, soit $d'(U)$ le $\mathcal{O}_S$ -module des sections de $\mathcal{F}^*$ sur U . Par (EGA, II, 6.3.4), les $d'(U)$ pour U variable forment un $\mathcal{O}_S$ -module quasi-cohérent d' sur $\mathcal{F}$ qui sera appelé le $\mathcal{O}_S$ -module des sections de $\mathcal{F}^*$ sur $\mathcal{F}$. Soit $\mathcal{F}^*$ soit un $\mathcal{O}_S$ -algèbre représentable et affine. Le champ algébrique qui est associé par (4.20) au $\mathcal{O}_S$ -module d' sera appelé la normalisation de $\mathcal{F}$ par rapport à $\mathcal{F}^*$. Sa formation est compatible avec tout changement de base étale.

Théorème (4.21). — Soit $\mathcal{F}$ soit un champ quasi-séparé sur un schéma noethérien S .

Supposons que (i) le morphisme diagonal $\mathcal{F} \rightarrow \mathcal{F} \times \mathcal{F}$ est représentable et non ramifié; (ii) il existe un schéma X de type fini sur S et un morphisme lisse et surjectif $X \rightarrow \mathcal{F}$. Alors, $\mathcal{F}$ est un champ algébrique de type fini sur S . M. Artin a développé des méthodes puissantes pour relier la représentabilité d'un champ à l'existence de morphismes étales surjectifs $x : X \rightarrow \mathcal{F}$ § 5. Deuxième preuve du théorème d'irréductibilité. Soit $\mathcal{C}_g$ ($g \geq 2$) le champ dont la catégorie des sections sur un schéma S est la catégorie des courbes stables de genre g sur S , les morphismes étant les isomorphismes de schémas sur S . Par (I. ii), le morphisme diagonal $\mathcal{C}_g \rightarrow \mathcal{C}_g \times \mathcal{C}_g$ est représentable, de type fini et non ramifié. Nous avons vu dans [1] que le champ classifiant les courbes stables de genre g est représenté par un schéma H_g , lisse et de type fini sur $\text{Spec}(\mathbb{Z})$. Le "oubli" morphisme $H_g \rightarrow \mathcal{C}_g$ est représentable, lisse et surjectif. En effet, si $p : C \rightarrow S$ est une courbe stable sur S , définissant un morphisme $\sim : S \rightarrow \mathcal{C}_g$, alors le produit fibré $\text{Fig} \times_S C$ est un schéma, lisse sur S , d'isomorphismes entre l'espace projectif standard de dimension $3g-6$ sur S et $\mathbb{P}(p_*(\mathcal{O}_C(-c)))$. Nous déduisons de ceci et (4.21) que :

Proposition (5.1). — $\mathcal{C}_g$ est un champ algébrique séparé de type fini sur $\text{Spec}(\mathbb{Z})$.

Soit us denoter par $\mathcal{C}_g$ le ouvert de $\mathcal{C}_g$ qui "consiste de" courbes lisses, et par $\mathcal{C}_g^*$ la "courbe universelle" sur $\mathcal{C}_g$, le champ algébrique classifiant les courbes stables pointées.

Théorème (5.2). — Les champs algébriques $\mathcal{C}_g$ et $\mathcal{C}_g^*$ sont propres et lisses sur $\text{Spec}(\mathbb{Z})$.

et le complémentaire de $\mathcal{C}_g^*$ dans $\mathcal{C}_g$ est un diviseur à croisements normaux relatif à $\text{Spec}(\mathbb{Z})$.

Définition (5.4). — Une structure de $\mathcal{O}_S$ -module de niveau k sur X est un isomorphisme (respectant leur

structure symplectique homogène) entre $\text{Rlp}_k(X/kZ)$ et $(Z/kZ)^{2g}$. (5.5) Étant donné une section s de p , et un ensemble de nombres premiers P incluant toutes les caractéristiques résiduelles de S , le théorème de spécialisation pour le groupe fondamental permet de construire un objet pro- $\text{tel}(X/S, s)/P$ de la catégorie (l.c. gr (P)) des faisceaux localement constants de groupes finis d'ordre premier à P , avec les propriétés (i) $H^0_{\text{ms}}(\sim^1(X/S, s)(P), G) = \text{Rlp}_k(X \bmod s, p^*G) = \text{Rlp}_k(\text{Ker}(p^*G \rightarrow s, G))$ fonctuellement en G (l.c. gr a') ; (ii) la formation de $\text{hi}(X/S, s)(P)$ est compatible avec tout changement de base.

io6 P. DELIGNE AND D. MUMFORD Si G et H sont deux faisceaux de groupes sur Set , nous définissons le faisceau des morphismes extérieurs de H à G , $H^0_{\text{me}}(H, G)$, comme le quotient de $H^0_{\text{m}}(H, G)$

by the action of H induced by its action on itself by inner automorphisms. The sheaf $H^0(\mathcal{O}_S(p^*G))$ is "independent" of the choice of s . We shall denote it as $\text{Uomyt}(\text{rzl}(X/S)(P), G)$. As $p : X \rightarrow S$ admits sections locally for the étale topology, this sheaf makes sense without assuming p to have a global section. It is independent of the choice of P , so long as P is prime to the order of G and includes all residue characteristics of S .

Définition (5.6). — Soit G soit un fini group of order n prime to P . A Teichmüller structure of level G on X is

covering of X^0 and $X_1 = S \times_{\mathcal{M}_g} (G \cdot \mathcal{M}_g)$ is the normalisation of X with respect to this covering of X^0 .

By the explicit local description (5.10), for any prime number l prime to n , $X_1 \times \text{Spec}(\overline{\mathbb{F}}_l)$ is the normalisation of $X \times \text{Spec}(\overline{\mathbb{F}}_l)$ with respect to $X_1 \times \text{Spec}(\overline{\mathbb{F}}_l)$. As this is true for any modular family, we get (5.9).

Corollary (5.11). — The geometric fibres of the projection

$$G \cdot \mathcal{M}_g^0 \rightarrow \text{Spec}\left(\mathbb{Z}\left[\frac{1}{n}\right]\right)$$

all have the same number of connected components.

$$\text{Aut}^0(\Pi) \rightarrow \mathbf{Sp}_{2g}(\mathbb{Z}),$$

is surjective (see [Ma]).

Theorem (5.13). — The number of connected components of any geometric fibre of the projection of $G \cdot \mathcal{M}_g^0$ onto $\text{Spec}(\mathbb{Z}[1/n])$ is equal to the number of orbits of $\text{Aut}^0(\Pi)$ in the set of exterior epimorphisms from Π to G .

Proof. — By (5.13), it suffices to prove that $G \cdot \mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})$ has the said number of connected components. As results from (4.14),

$$\pi_0(G \cdot \mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})) = \pi_0({}_G M_g)$$

where ${}_G M_g$ is the coarse moduli scheme classifying stable smooth curves of genus g over $\mathbb{C}$

$$H_1(\mathbb{C}, \mathbb{Z}) \sim \pi_1(\mathbb{C}) / \langle \pi_1(\mathbb{C}), \pi_1(\mathbb{C}) \rangle$$

and $\Pi / \langle \Pi, \Pi \rangle$. By Teichmüller's theory [W], the analytic space T_g classifying Teichmüller curves of a given genus $g \geq 2$ is homeomorphic to a ball, and hence connected.

If ψ is a surjective homomorphism from Π to G , the map

$$(C, \varphi) \mapsto (C, \psi\varphi)$$

defines a morphism $t_\psi : T_g \rightarrow {}_G M_g$. Two such maps t_ψ and $t_{\psi'}$ have the same image if and only if $\psi = \psi'\sigma$ for $\sigma \in \text{Aut}^0(\Pi)$, and

$${}_G M_g = \coprod_{\psi \bmod \text{Aut}^0(\Pi)} t_\psi(T_g)$$

which implies (5.13).

" $\mu(\varphi)$ of φ by the commutative diagram

$$\begin{array}{ccc} \wedge^2(\mathbf{Z}/n\mathbf{Z})^{2g} & \longrightarrow & \mathbf{Z}/n\mathbf{Z} \\ \downarrow \wedge^2 \varphi & & \downarrow \mu(\varphi) \\ \wedge^2 \mathbf{R}^1 p_* (\mathbf{Z}/n\mathbf{Z}) & \xrightarrow{\wedge} & \mu_n^{\otimes -1} \sim \mathbf{R}^2 p_* (\mathbf{Z}/n\mathbf{Z}) \end{array}$$

of isomorphism between $\mathbf{Z}/n\mathbf{Z}$ and $\mu_n^{\otimes -1}$ is $\mathrm{Spec}(\mathbf{Z})$

a surjective exterior homomorphism from $\mathrm{nl}(X/S)$ to G . The finite generation of $\mathrm{zI}(X/S)$ (SGA, 6o/6I, exp. IO) implies:

Lemme (5.7). — The sheaf on (Sch/S) of the Teichmüller structures of level G on X is

represented by an étale covering of S . We denote by $\sim, 0$ the champ classifying the stable lisse curves of genus g and G g characteristic prime to n , with a Teichmüller structure of level G . For any algebraic champ dr' , we denote by $\sim s$ its ouvert $\mathrm{dr}' \times \mathrm{Spec}(\mathbf{Z}[i/n])$.

Lemme (5.7) may now be rephrased:

Proposition (5.8). — The "forgetful" morphism

is representable, finite and étale. The champ $a.A'$ is thus an algebraic champ. Soit $e \sim go$ be the normalisation of $g[0[I/n]]$ with respect to $a.g \sim$. The champ $o.g0$, being representable and finite over $g0[I/n]$, is proper over $\mathrm{Spec}(\mathbf{Z}[i/n])$.

Théorème (5.9). — The geometric fibres of the projection of $a.go$ onto $\mathrm{Spec}(\mathbf{Z}[i/n])$ are

normal, and, fibre by fibre, $\sim.g \sim$ is dense in $\sim.go$. Démonstration. — We will use Abhyankar-Artin's lemma, in its "absolute" form:

Lemme (5. xe). — Soit D soit un divisor with normal crossings on an excellent regular

schéma X , Y an étale covering of $X - D$ and Y the normalisation of X with respect to Y . Assume that the generic points of the irreducible components of D are all of characteristic o . Then every geometric point of X has an étale neighbourhood $x : X' \rightarrow X$ such that, on X' : (i) D becomes a union of regular divisors (D_i) , $i=1, \dots, r$, of equation $t_i \simeq 0$. (ii) There exists an integer k prime to residue characteristics of X' such that Y becomes isomorphic to a disjoint union of quotients (by subgroups of g_i) of the covering of X' obtained by extracting the k -th roots of the t_i 's. If $x : X \rightarrow g[g[I]]$ is an étale morphism, then $X \sim$ is the complement $L \subset X$ of a divisor with normal crossing relative to $\mathrm{Spec}(\mathbf{Z})$, $X \simeq L \times (\mathrm{aMl} \sim g)$ is an étale

Corollaire (5. xI). — The fibres géométriques of the projection

$G \rightarrow \text{Spec } Z$ all have the same number of connexes composantes. Démonstration. — By (4.17), all fibres géométriques $\text{odt}'gX \text{Spec}(F_z)$ of $a \sim t'g$ over $\text{Spec}(Z[I/n])$ have the same number of connexes composantes. These connexes composantes are irréductible (4- 16). Furthermore $a \sim \Omega_X \text{Spec}(F_z)$ is dense in $\text{adt}'g \times \text{Spec}(F \sim)$ and thus has the same number of connexes (or irréductible) composantes as $Q.A \sim g \times \text{Spec}(F_z)$, and this number is independent of I . (5- x2) Soit us denote by H the fundamental group of an oriented closed differentiable surface SO of genus g . The group Π may be defined by generators and relations as follows: (i) generators: x_i for $i < i < 2g$; (ii) relation: $(x \sim, x_{a+1}) \dots (x \sim, x_{a+ \sim}) \dots (x_g, x_{2a}) = e$, where $(a, b) = aba^{-1}b^{-1}$. Soit $(e \sim)$ be the standard basis of $Z2g$. The morphisme φ from H to Zeg with $\varphi(X_i) = e_i$ identifies $H/(H, n) = H_1(n, Z)$ with $Z \sim a$. The surface SO is a $K(\Pi, i)$ and thus $H^0(H, Z) = I-P(S_0, Z) = Z$, and the cup product defines a symplectic structure on $\Pi/(\Pi, H)$. This structure is identified by φ with the standard symplectic structure of $Z2g$. We denote by $\text{Aut} \sim$ the subgroup of $\text{Aut}(\Pi)$ which acts trivially on $H_2(\Pi, Z)$. Dehn has proved that each exterior automorphisme of Π is induced by a diffeomorphisme of SO onto itself and that the map induced by φ : $\text{Aut} \sim \rightarrow \text{Sp} \sim(Z)$, is surjectif (see[Ma]).

Théorème (5. I3). — The number of connexes composantes of any fibre géométrique of the

projection of $Q \sim o$ onto $\text{Spec}(Z[I/n])$ is equal to the number of orbits of $\text{Aut} \sim$ in the set of exterior epimorphisms from H to G . Démonstration. By (5. I3), it suffices to prove that $0 \sim G \text{dt}'gx \text{Spec}(C)$ has the said number of connexes composantes. As results from (4-14), $\text{rr}_0(a \sim t' \sim X \text{Spec}(C)) = r:0(GM_g)$ where aM_g is the coarse moduli schéma classifying stable lisse curves of genus g over C with a Teichmüller structure of level G .

$\sim o8$ P. DELIGNE AND D. MUMFORD Recall that a Teichmüller π urve of genus g is a stable lisse curve C of genus g over $(I$ together with an exterior isomorphisme φ of the transcendental fundamental group $\text{rq}(C)$ with H , which induces a symplectic isomorphisme (1) between $HdC, Z) \sim \sim dC)/(\sim dC), \sim 1(C))$ and $\Pi/(\Pi, H)$. By Teichmüller's theory [W], the analytic space T_0 classifying Teichmüller curves of a given genus $g > 2$ is homeomorphic to a ball, and hence connexes. If $\sim \varphi$ is a surjectif homomorphisme from Π to G , the map $(C, 9) \sim (C, +9)$ defines a morphisme $t+ : T_0 \rightarrow GM_0$. Two such maps $t+$ and $t+$, have the same image if and only if $+ = + \sim$ for $\sim \text{Aut} \sim$ and $H t \sim, (T_0) GMa = r \text{ mod } \text{Aut} \sim$ which implies (5- i3). (5. x4) Soit us denote by $n \sim \sim$ the algebraic champ classifying stable lisse curves with a Jacobi structure of level n . This algebraic champ " is " a true schéma for $n > 3$ (by [S]). If 9 is a Jacobi structure of level n on a stable lisse curve $p : X \sim S$, we define the " multiplicator " $\sim (9)$ of 9 by the commutative diagram $A(z/nz)20 z/nz A2Rlp, (g/nZ) A > \sim \sim K2p, (g/ng)$ The schéma of isomorphismes between Z/nZ and ~ 1 is $\text{Spec}(Z[e \sim \sim j, i/n])$ thus $9 \sim \sim (9)$ defines a morphisme $\sim$ from S to $\text{Spec}(Z[e \sim \sim /, I/n])$. This being true for any X and S , $[z$ is induced by $: , dr' \sim \sim \text{Spec}(Z[e \sim /n, I/n])$.

Théorème (5. x5). — The geometric fibres of the morphisme $\sim$ are connexes.

Démonstration. $\sim$ By definition, $n \sim \sim$ is open and closed in $G \text{ Jr}'0 \sim$ for $G = (Z/nZ) \sim$. The group $GL2g(Z/nZ)$ acts on G , and thus on $\text{odt}' \sim$ One has: (i) the ouvert $,all \sim$ of $\sim /t' \sim$ is stable under the subgroup $H = CSp2g(Z/nZ)$ of symplectic similitudes;

(ii) $-J = \Pi o(de' tl 0 G/H \sim n gJ 9 (1)$ An arbitrary isomorphisme 9 induces an isomorphisme between $Hx(C, Z)$ and $\Pi/(\Pi, \Pi)$ which is always symplectic up to sign.

Lemme (5- x6). — The homomorphisme

$\text{Aut} \sim \rightarrow \text{Sp} \sim(Z/nZ)$ is surjectif. Démonstration. — This results from Dehn's theory (5. I2) and from the fact that the groups Sp_n , being split semi-simple simply connexes groups, are

generated by their unipotent elements, so that the reduction map $\mathrm{sp} / \mathbb{Z} \rightarrow \mathrm{Sp} / \mathbb{Z} / \mathbb{Z}$ is surjectif.

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